

**DEPARTMENT OF STATE
DRINKING WATER SAFETY PROGRAM**



**Prepared by:
Bureau of Overseas Building Operations
(OBO/OPS/SHEM and OBO/OPS/FAC)**

and

**Bureau of Medical Services
(MED)**

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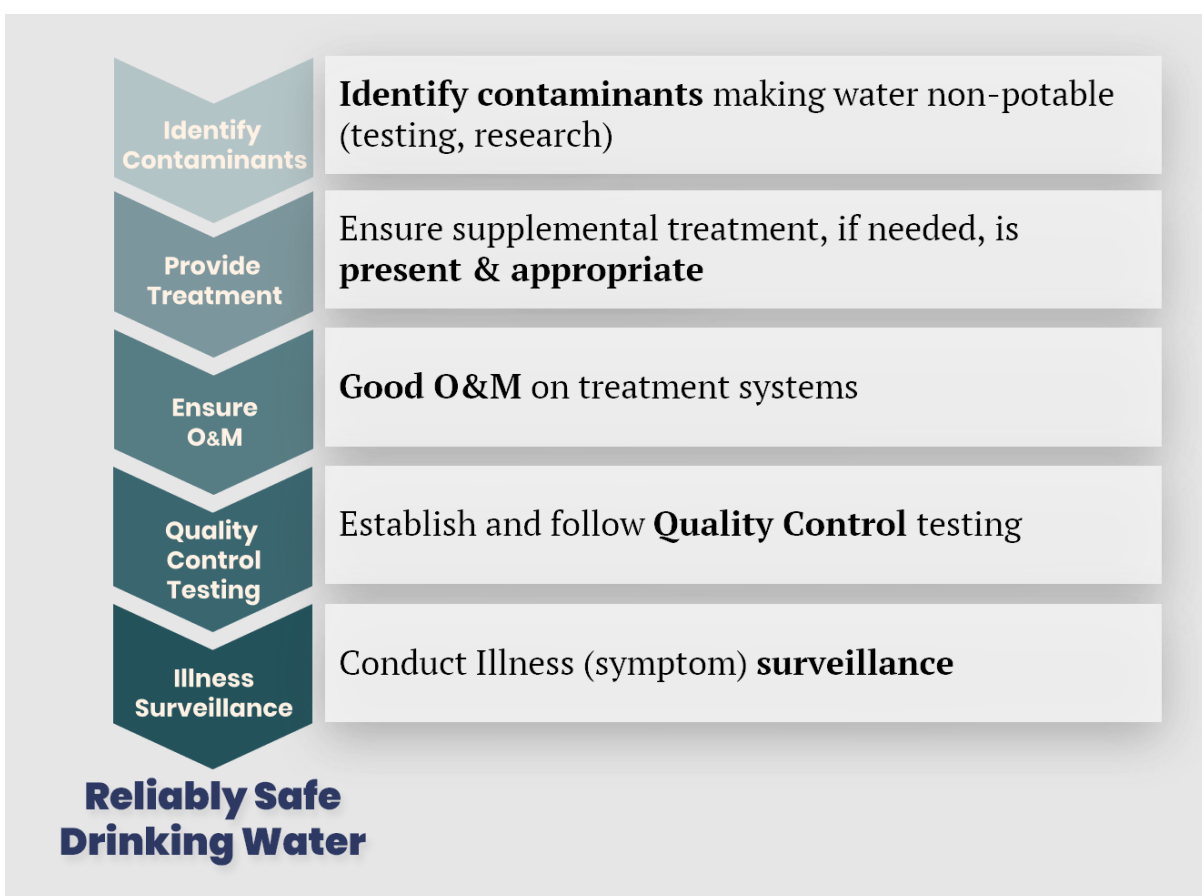
Version History

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1. EXECUTIVE SUMMARY

Per 15 FAM 957.5 Drinking Water Safety Program Requirements¹: *‘Posts must provide potable² water that is microbiologically and chemically safe to drink at all Department of State-owned or -leased facilities abroad where employees and post family members work or reside. Posts will accomplish this in part by periodically testing the water; ensuring that appropriate water treatment devices or systems are provided, used, and properly maintained; and ensuring that any bottled water provided is safe to drink.*

To help posts meet their responsibility to ensure safe drinking water is available to employees and their families, the following elements should be addressed and implemented:



¹ 15 FAM 957.5, CT:OBO-101; 06-24-2020

² “Fit for Human Consumption” i.e ‘potable water’ is water that meets primary drinking water standards as defined by U.S. Federal Safe Drinking Water Act (SDWA) of 1974 and its amendments of 1986 and 1996 (Title 42 U.S. Code, Subchapter XII – Safety of Public Water Systems, §300f-j and is safe for drinking, cooking, bathing, showering, dishwashing, and maintaining oral hygiene.

This is established through development and implementation of individual Post Drinking Water Safety Programs (DWSPs), which include the following elements:

- Evaluating and certifying the quality of the source water,
- Coordinating with OBO on appropriate potable water treatment system equipment (should treatment be required),
- Implementing appropriate operations and maintenance (O&M) procedures for the potable water treatment systems,
- Implementing a quality control (QC) program to verify that water quality remains reliably safe for human consumption (potable),
- Providing bottled drinking water after other means of treatment have been determined to be not appropriate and not cost-effective,
- Conducting disease surveillance potentially caused by exposure to unsafe drinking water, and
- Determining whether children at post need fluoride supplementation in accordance with MED's supplementation criteria.

This document describes Department of State (DoS) policy and guidance used to address these elements and meet the requirements of 15 FAM 957.5. Included are DoS standards for drinking water, guidance for establishing individual Post Drinking Water Safety Programs (DWSPs), and a summary of roles and responsibilities between Post, the Bureau of Overseas Building Operations (OBO), and the Bureau of Medical Services (MED) to support, fund, and oversee the DWSPs for each Post.

2. BACKGROUND

Policy and Purpose

Providing water that is microbiologically and chemically safe (i.e. safe for human consumption, or 'potable') per 15 FAM 957.5 Drinking Water Safety Program Requirements, is a shared responsibility between Posts, OBO, and MED. Posts have the ultimate responsibility to develop and implement a Drinking Water Safety Program (DWSP) that is designed to control water contaminants of greatest health concern for their local water sources, and any treatment, storage, and distribution systems in place. OBO provides standards, guidance, verification, auditing, and funding for the Posts' DWSP programs. MED conducts health surveillance, coordination on water-borne disease outbreaks, implements the MED fluoride supplementation program, and provides on-site support on water quality issues.

Each Post's DWSP should describe the system and procedures in place to provide safe water on an ongoing basis and provide information on fluoride levels for the MED fluoride supplementation program. Implementation of the policies and guidance provided in this document will allow Posts to fulfill the requirements in 15 FAM 957.5.

Comments and suggestions to improve this guidance are welcome at any time. A DoS Drinking Water Safety Group will be established to develop the process for updating this document. This Group will be comprised of representatives from:

- OBO/OPS/SHEM for drinking water safety, potability, and health related issues,
- OBO/OPS/FAC/PS for operations and maintenance of potable water treatment systems and equipment, and
- OBO/PDCS/DE for design standards and engineering, design, and technical reviews of potable water treatment systems,
- OBO/CFSM/CM for construction and commissioning of NEC and NCC potable water treatment systems,
- MED for disease surveillance potentially related to water contamination, and fluoride supplementation

Questions on the process for developing this document should be directed to the Office of Safety, Health, and Environmental Management (OBO/OPS/SHEM)³ and Office of Facility Management (OBO/OPS/FAC/PS)⁴.

Effective Date

11/1/2023 (Version 1)

3. DEPARTMENT DRINKING WATER STANDARDS

Department drinking water quality standards are based on whether the water used for human consumption is:

- 1) Post provided/distributed (as treated or untreated city water, well water, trucked water, atmospherically generated water, and/or internally bottled water), **OR**
- 2) Commercially bottled water.

Verification of the quality of water provided for human consumption is done by evaluating water quality testing results against these standards.

Standards for Post Provided/Distributed Drinking Water

The Department's standards for Post provided/distributed drinking water are based on the U.S.

³ For OBO/OPS/SHEM support, contact opmsj@state.gov

⁴ For OBO/OPS/FAC/PS support, contact Water_Testing_Group@state.gov

Environmental Protection Agency (EPA) National Primary Drinking Water Regulations⁵ (NPDWR) as well as the National Secondary Drinking Water Standards (NSDWR). Use of these standards is codified in the Department's OBO Plumbing Code⁶ and standard specifications for potable water treatment design and construction.⁷ The EPA Primary Standards (NPDWRs) are developed based on health effects of various parameters and the Secondary Standards (NSDWRs) are based on aesthetic effects (non-health related). ***These EPA Primary and Secondary Drinking Water Standards are found in Appendix A.***

The US EPA Primary Drinking Water Standards (NPDWRs), fall into five (5) categories of water contaminants that are of concern when assessing safe drinking water:

- Microorganisms;
- Disinfectants and disinfectant by-products;
- Inorganic chemicals;
- Organic chemicals; and
- Radionuclides.

These contaminants can further be divided into 'Acute' contaminants and 'Chronic' contaminants.

'Acute' contaminants can cause short-term health effects within hours or days of exposure. **Microorganisms are the primary concern with drinking water quality since they can cause acute and widespread health effects that are often severe and immediate.** Although water is rarely sterile, minimizing the presence of microorganisms and the absence of the most hazardous (such as those resulting from fecal contamination) will be most protective of human health.

'Chronic' contaminants can cause health effects after continuous long-term exposure at levels greater than the maximum contaminant levels (MCLs) regulated by the USEPA. **There is no immediate health risk from consuming drinking water containing a regulated chronic contaminant at levels below the MCL.** However, sensitive populations (the very young, elderly, pregnant women, and cancer patients) may be more susceptible to adverse health effects from chronic contaminants. Examples of 'chronic' drinking water contaminant include inorganic and organic chemicals (including disinfectants and disinfection-by-products) and radionuclides.

⁵ 40 CFR Parts 141-142 and the Safe Drinking Water Act (PL 93-523, PL 104-182, etc.) as amended

⁶ OBO-ICD 2020 OBO – International Codes Supplements, Amendments to the International Plumbing Code, January 2020 (Chapter 3, 312.11 and 312.12; Chapter 6, 601.4 (tests),

⁷ United States Department of State Overseas Buildings Operations OBO Standard Specifications, Section 223230 – Potable Water Treatment. January 2018R.

The Department's policy is to control both acute and chronic parameters in water provided for human consumption - by utilizing OBO standards for design, construction, and commissioning for potable water treatment systems; supporting Post's on-going operations and maintenance of their drinking water systems (treatment, storage, and distribution); and by establishing standards and criteria for use of certified commercially bottled water.

Standards for Commercially Bottled Drinking Water

Department standards for commercially bottled water (when considered necessary) require that the bottled water be:

- Certified by the NSF International (NSF), www.nsf.org⁸, or
- From a reliable source (e.g. US military certified, or endorsed by the International Bottled Water Association (IBWA) or UL Certified).

The United States Food and Drug Administration (USFDA) has developed regulations and standards for bottled water which are used as the basis for certification by the IBWA⁹ and UL¹⁰. ***These Bottled Water Standards are found in Appendix A.*** The USFDA regulations include 'Current Good Manufacturing Practices (CGMPs) for the processing and bottling of bottled drinking water. These include that it be processed, bottled, held, and transported under sanitary conditions, and found to be safe and sanitary through sampling and analysis. These regulations also require proper plant and equipment design, bottling procedures, and record keeping."

The USFDA has not established a 'shelf life' for bottled water¹¹. IBWA advises consumers to store bottled water at room temperature (or cooler), out of direct sunlight, and away from solvents and chemicals such as gasoline, paint thinners and dry cleaning chemicals. Commercially Bottled water can be used indefinitely if it meets the potability requirements in this policy and is stored properly.

Standards for MED Fluoride Supplementation Program

The MED Fluoride Supplementation program provides guidance to local Health Units on fluoride

⁸ NSF bottled water certification: <https://www.nsf.org/testing/food/food-beverage-product-certification/beverages-packaged-ice-certification>

⁹ IBWA. Bottled Water Code of Practice. Revised November 2020, Effective January 1, 2021. International Bottled Water Association. Alexandria, VA. <https://www.bottledwater.org>.

¹⁰ USFDA: The United States Food and Drug Administration, Code of Federal Regulations – Beverages and Bottled Water: Chapter I, Title 21, Part 165, Subpart B, Section 165.110. <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?fr=165.110#:~:text=Sec.,safe%20and%20suitable%20antimicrobial%20agents>.

¹¹ <https://www.fda.gov/consumers/consumer-updates/bottled-water-everywhere-keeping-it-safe>

supplementation for Department employees and their families overseas and is implemented by Post's Health Units. Fluoride supplementation is based in part on fluoride levels measured in the drinking water, and therefore may be included in the testing program described in a Posts DWSP. ***A description of the MED Fluoride Supplementation program can be found in Appendix B.***

4. DRINKING WATER SAFETY PROGRAM (DWSP) GUIDANCE

Post Drinking Water Safety Programs (DWSPs) should be designed to guide ongoing system management and operations to ensure continued reliability and safety of the water provided for human consumption. The key elements of any DWSP should be:

- Identifying a Post DWSP team,
- Describing and verifying the system to provide drinking water fit for human consumption,
- Establishing management and operations procedures for both normal operations and incident situations,
- Verifying the DWSP through quality control monitoring and auditing,
- Reviewing and updating the DWSP.

These elements are described below and are based on a risk assessment/risk management approach to address potential contaminants of greatest concern. These elements should be described in a Post specific DWSP document that can be used to guide management, operations, and communications related to provision of safe drinking water. This is a 'living' document that is reviewed annually and updated as needed to reflect changes in source water, treatment systems, staffing and responsibilities at Post. ***Appendix C contains an example DWSP Table of Contents that Posts should use to develop a Post specific DWSP.***

Identifying a DWSP Team

The DWSP Team should be identified by the Management Officer and include key personnel from Management, Facilities, and the Health Unit that have responsibility for management, operations, testing, verification, and communications related to providing safe drinking water at Post. At a minimum, the DWSP Team should include:

- Management Officer
- POSHO
- Facility Manager
- Health Unit Medical Officer and/or Regional Medical Officer
- Regional Medical Laboratory Specialist (RMLS)

The team should be assembled to develop the initial DWSP for Post, and then meet on a regular basis to review status of the program and incorporate any changes needed.

Describing and Verifying the System

The DWSP should describe the current system in place for providing water safe for human consumption. This description should include information on where the water is coming from (source), an assessment of the quality of that water, and a description of any water treatment, storage, and distribution systems in place.

Source Water Description

The source of water used for human consumption may be the city or local water utility, Post surface water or groundwater wells, commercially trucked water, commercially bottled water, atmospheric water generators – or any combination of these sources. The type of source water from the city or local water utility could be surface water or ground water and could be treated or untreated prior to arriving at Post. Post should clearly identify the type of source water in the DWSP, including the contact names/numbers for the City or Local Water Utility, trucked water, and commercially bottled water sources.

Source Water Assessment and Justification for Bottled Water

Source Water Assessment. Assessment of the source water quality is a shared responsibility between Post and SHEM. ***Department guidelines and procedures for source water assessments are described in Appendix D.*** The specific requirements for frequency of Source Water Assessment depend on the type of source water:

Source Water Type	Minimum Frequency	Responsibility
Surface Water	every 5 years	OBO (OBO/OPS/SHEM)
Groundwater	Every 5 years	OBO (OBO/OPS/ SHEM)
Commercially Trucked Water	Prior to issuing contract and at each delivery	Post with assistance from OBO (SHEM)
Commercially Bottled Water (Certified)	Prior to issuing contract	Post with assistance from OBO (SHEM)
Commercially Bottled Water (if not certified)	Prior to issuing contract then annually	Post with assistance from OBO (SHEM)

Surface or Ground Water obtained from a City or Local Water Utility, or directly from Post (i.e. on-site well) should be assessed every five years at a minimum, to evaluate contaminants of concern and review the host country's public health structure and the water provider's practices and performance. **This assessment is completed by OBO/OPS/SHEM with assistance from Post.**

Commercially Trucked Water should be assessed prior to contracting with the supplier, and then annually at a minimum. **This assessment can be completed by Post, with OBO/SHEM support.** Water quality should be verified at each delivery as part of the Posts operational monitoring/quality control (OpMon/QC) testing plan.

Commercially Bottled Water company that is not certified should be assessed prior to entering into a contract with a supplier, then annually at a minimum, to verify quality. **This assessment can be completed by Post with OBO/SHEM support.**

Justification for Bottled Water. Post may use commercially bottled water when water treatment options are not feasible or while treatment devices are being installed or repaired. This can be expensive and will require justification and a cost-benefit analysis. If commercial bottled water is warranted, SHEM will prepare a memo for the post FMO that provides justification in accordance with the requirements outlined in 4 FAM 447.3. This justification is based on the source water quality assessment and a cost comparison of treatment versus bottled water. SHEM will work with Post to develop the information required. ***An example of typical treatment versus bottled water cost comparison is provided in Appendix E.***

Post Treatment, Storage, and Distribution Systems

If the source water **DOES NOT** meet Department standards, or distribution system deficiencies or water storage facilities degrade the quality of the source water, then treatment may be installed to provide safe drinking water.¹² This can be done by installing point of entry (POE) water treatment systems that treat all the water at the point of entry to a building or compound, and/or installing point of use (POU) water treatment devices that treat water at the tap. Proper selection of treatment equipment and devices, operating and maintaining them, and testing the quality of the water produced is key to providing safe drinking water on an ongoing basis. ***Appendix F contains background on water treatment systems and provides guidance for selection and installation.***

If the source water does not need to be treated to meet Department standards, there may still be water storage facilities (water storage tanks, cisterns) and distribution systems in place (piping networks or internally bottled water distribution) to provide water to the consumer. ***The quality of the water can degrade during storage and distribution, so the Post DWSP should include a description of all storage and distribution systems in place.***

Establishing Management and Operations Procedures

The Post DWSP should describe procedures in place for both normal operations and incident situations to ensure ongoing provision of safe drinking water. These procedures should address management, operations and maintenance, and communications. ***Appendices G and H provide guidance on procedures for both routine O&M and incident responses.***

Conducting on-going operational water quality monitoring is key to determining proper operation of any treatment equipment. All posts that manage POE potable water treatment equipment or POU water treatment devices should have a written Operational Water Quality monitoring plan in place. The Operational Water Quality monitoring plan should be Post specific and based on the source water quality and treatment systems in place. In general, it will include any primary regulated parameters

¹² OBO-ICS 2022 OBO-International Codes Supplements, Amendments to the International Plumbing Code, January 2022 (Section 611 Drinking Water Treatment Units)

the system is designed to remove, as well as secondary regulated parameters of concern and parameters that indicate corrosiveness of the water. The Operational Monitoring Plan should be combined with a Quality Control (QC) monitoring plan (described in the next section). This combined Operational Monitoring/Quality Control (OpMon/QC) plan will provide information on proper O&M of treatment equipment (operational monitoring) as well as verification of the safety of the water for human consumption (potability) (QC monitoring). ***OBO/OPS/FAC/PS and OBO/OPS/SHEM will work with Post to develop their Operational Monitoring/Quality Control Testing Plan (OpMon/QC Test Plan) for inclusion in their DWSP.***

OBO has implemented a Certificate to Operate (CTO) Program for Potable Water Treatment Systems at Posts and Missions. The purpose of the CTO program is to assist Posts in ensuring their Potable Water Treatment equipment and operations and maintenance activities produces water that is fit for human consumption (potable) on an ongoing basis. This program is implemented by OBO/OPS/FAC (FAC Water Testing Group) in coordination with OBO/OPS/SHEM and MED. The program is voluntary, and involves a sanitary survey/site inspection, identification of any deficiencies in the system, and a commitment by Post to work with OBO and MED to: 1) correct deficiencies, 2) implement an OpMon/QC testing plan, and 3) agree to ongoing evaluation of whether the water is safe for human consumption (potable). Guidance for the Potable Water CTO Program is available through OBO/FAC Water Testing Group at Water_Testing_Group@state.gov. Posts involved in the CTO program may reference their participation in their DWSP.

Verifying the DWSP Through Quality Control Monitoring and Auditing

Verifying that drinking water at posts around the world is safe for human consumption (potable) is not a simple process and requires evaluating a combination of Quality Control (QC) monitoring results and disease surveillance data.

Quality Control (QC) Monitoring

Quality Control (QC) Monitoring is periodic testing of the drinking water to ensure that the water is reliably safe for human consumption (potable). This may include QC monitoring at Posts with supplemental treatment systems (POE and POU systems), as well as monitoring of uncertified bottled water and trucked water. The frequency and scope of monitoring is not universal and depends upon the contaminants identified in the source water assessment. Each post should have a written QC monitoring plan. For Posts that operate POE potable water treatment equipment and/or POU water treatment devices, the QC monitoring plan should be combined with the Operational monitoring plan (OpMon/QC Plans). SHEM evaluates the QC monitoring data to determine potability (i.e. if the water is safe for human consumption) (See Appendix D).

Microbiological Testing. QC Monitoring Plans focus on microbiological quality of the water because of the acute nature of these contaminants and the ability to test for them on an ongoing basis to

determine safety for human consumption (potability) and proper operation of treatment equipment. The three most important parameters to determine if water is reliably safe from microbiological contaminants are:

- Total Coliform/E.Coli bacteria
- Turbidity
- Free Chlorine Residual

These parameters are referred to as ‘the Big Three’ and should be included in any QC Monitoring Plan for posts receiving water directly from a local utility or well and at Posts operating supplemental treatment with chlorination. These parameters should be tested on an ongoing basis as identified in the QC Monitoring Plan (weekly or monthly or quarterly). Samples can be collected and analyzed on-site by Post personnel or by an outside contractor/laboratory. If a Post specific QC Monitoring Plan is not yet in place, post should sample at 2 or 10% (whichever is greater) of drinking water taps or uncertified bottles of water, and analyze the samples for total coliform, E. Coli, and turbidity. If water is chlorinated, they should also measure free chlorine residual in the samples, using the following criteria:

If the finished water is....	Monitor for....	Minimum Frequency	Quality Standard
Chlorinated	Free chlorine	Monthly	0.2 – 1.00 mg/L
Unchlorinated	Total Coliform bacteria	Monthly	Negative
	E.coli, if coliform positive		
Uncertified Bottled water or Bottled water bottled by post	Total Coliform bacteria	Quarterly (monthly if post is bottling water)	Negative
	E. coli if coliform positive		
NSF certified bottled water US military certified bottled water	No testing required		

Chemical Testing. It would be impractical to test for all the USEPA regulated chemical contaminants at the same frequency as microbiological contaminants. These chemical contaminants are controlled by selection of proper treatment equipment and/or use of certified bottled water. **However, lead, nitrates, and arsenic are three water parameters that may merit specific verification after microbiological concerns, as they may be present at high enough**

concentrations to cause immediate health effects. Consumption of water containing elevated lead or nitrate levels can pose a health risk to children and arsenic causes skin damage, problems with circulatory system, and may increase the risk of getting cancer. Table 1 provides recommendations for frequency of QC monitoring for these contaminants, which should be analyzed by a certified laboratory.

Table 1. QC Testing for Lead, Nitrates, Arsenic*

Parameter and Regulated Level	Where to Test	Comments on Testing	Baseline Testing Frequency	Responsibility
Lead (Pb) No safe level of lead Standing sample Action Level = 0.015 mg/L)	<u>Compounds and Residences:</u> Kitchen taps and Source water*	Use OBO approved labs, sampling, and testing procedures. <u>Source water sampling procedures:</u> Flushed sample representative of the source water. <u>Kitchen tap sampling procedures:</u> 1 liter standing sample collected at kitchen tap. Faucet must not be used for 6 hours prior to sampling. Do not remove aerator.	Source water sampling: every 3 Years Standing sample at kitchen tap: To be determined Standing sample recommended if source water sample is above 0.005 mg/L and/or conditions warrant evaluating stagnation lead levels due to presence of leaded materials or previous elevated lead levels)	OBO (OBO/OPS/SHEM) will approve labs, sampling, and testing procedures. Post will conduct testing and report results.
Nitrate as nitrogen (NO₃-N) MCL = 10 mg/L	<u>Compounds and percentage of Residences:</u> Source water*	Use OBO approved labs, sampling, and testing procedures. Depending on the source, sample results may fluctuate seasonally.	<u>Compound:</u> Quarterly (minimum) if source water assessment is above the MCL . <u>Residences:</u> Annually (minimum) if source water assessment is above the MCL	OBO (OBO/OPS/OPS/SHEM) will approve labs, sampling, and testing procedures. Post will conduct testing and report results.
Arsenic MCL = 0.01 mg/L	<u>Compounds and percentage of Residences:</u> Source	Use OBO approved labs, sampling, and testing procedures. Depending on the source, sample results may fluctuate seasonally.	<u>Compound:</u> Quarterly (minimum) if source water assessment is above the MCL .	OBO (OBO/OPS/OPS/SHEM) will approve labs, sampling, and

Parameter and Regulated Level	Where to Test	Comments on Testing	Baseline Testing Frequency	Responsibility
	water*		Residences: Annually (minimum) if source water assessment is above the MCL	testing procedures. Post will conduct testing and report results.

*If water treatment is installed to specifically remove these compounds, OpMon/QC testing plans should include testing treated water for these compounds.

Disinfection by-products are compounds that may form when disinfecting water with chlorine. They can be present in the source water received from a local utility and be created during chlorine disinfection treatment and water storage at Post. Since many DoS potable water treatment systems use chlorination; disinfection byproducts are a potential contaminant of concern and should be part of the OpMon/QC testing. These compounds are chronic contaminants and do not cause an immediate risk to health, but because of their potential to be present, the Department **MAY** recommend ongoing QC monitoring for verification of levels as shown in Table 2.

Testing for Other Regulated Parameters. QC Monitoring for additional regulated parameters (inorganics, organics, radionuclides – See Appendix A) should be done at the frequency shown in Table 3.

Table 2. Disinfection By-Products (DPBs)

Parameter and MCL	Where to Test	Comments on Testing	Baseline Testing Frequency	Responsibility
Total Trihalomethanes (TTHM)* 0.08 mg/L	At the source and furthest tap in the Post distribution system	By-product of chlorine disinfection. Standard is the sum of all TTHMs or HAAs as an annual average of quarterly tests.	Quarterly testing once every 5 years. If average of samples is above the MCL, then quarterly testing every year.	OBO (OBO/OPS/OPS/SHEM) will approve labs, sampling, and testing procedures.
Haloacetic Acids (HAA)** 0.06 mg/L	Percentage of residential properties	Samples are time sensitive.		Post will conduct testing and report results.

*TTHMs include chloroform, bromoform, bromo-dichloro-methane, dibromo-chloro methane

**HAAs include dibromoacetic acid, dichloroacetic acid, monobromoacetic acid, monochloroacetic acid, and trichloroacetic acid

Table 3. Other Regulated Parameters

Parameter*	Where to Test	Comments on Testing**	Baseline Testing Interval
Inorganic Chemicals	Source water to compounds and residences	Analysis by an ISO 17025 certified lab using approved analytical methods	Once every 5 years unless samples are above the MCL, then annually
Organic Chemicals			
Radionuclides			
Fluoride			

*See Appendix A for list of chemicals and MCLs

**Contact OBO/OPS/SHEM or OBO/OPS/FAC/PS for guidance

Illness Surveillance

Post should actively monitor for diagnosed or positive cases of potential waterborne illness on Post, which may indicate a change in the source water or a failure of supplemental treatment. This surveillance should account for not just post population, but also the host country population, as this may provide warning about problems with the water utility.

Potability determinations

OBO/OPS/SHEM reviews and makes determinations about water potability for existing potable water treatment systems. The steps involved in making potability determinations may differ, depending on the situation or scenario, as initial source water assessments may require more research than a determination following changes or maintenance to water treatment systems. The processes for source water assessments and potability determinations are detailed in Appendix D.

Reviewing and Updating the DWSP

The DWSP should be reviewed annually by the DWSP team at Post. Any source, treatment, and property changes can be incorporated into the revised plan.

ROLES AND RESPONSIBILITIES BETWEEN POST, MED, AND OBO

	Management	Health Unit
POST	- Drinking Water Safety Plan (DWSP) - Appoint team - Develop and update plan - Oversee implementation - Facilitate coordination between Post FAC and HU - Coordinate communication on water safety - Post Community - Post FAC/HU - OBO, MED	- Disease surveillance Reporting - QC Testing (as determined in coordination with Post FAC) - Implementing the Fluoride Supplementation Plan

	Facilities • Co-development of Post DWSP • Water treatment system O&M • GMMS work orders • Operational Testing and coordination on QC Testing (with HU)	Facilities • Communication with Post Management and HU on: ○ Routine Operations ○ Water quality and operations Incidents ○ Changes to source and/or treatment systems for DWSP
MED	Headquarters • Illness surveillance, public health tracking • Communication and coordination with OBO	Regional Medical Laboratory Specialists • Consultation on water safety - Post • Coordination on water safety – OBO
OBO	PDCS • Design, construction, commissioning of NEC/NCC potable water treatment systems ○ Review and oversight of source water quality test results for NEC/NCC projects ○ Design appropriate for source water characteristics ○ Construction per OBO codes, standards, and specifications • Permits for rehab/replacement of potable water treatment systems • Source assessment to determine adequate system/supply resilience	OPS/SHEM • Source water assessments • Justification for bottled water • Potability determinations • Guidance ○ Incident response ○ Communications ○ Health and Safety ○ POU Treatment ○ Treatment selection and piloting
	CSFM/CM • Construction and commissioning of new potable water treatment systems per OBO codes, standards, and specifications	OPS/FAC • Post Support and Training ○ O&M for Potable Water Treatment Systems ○ Repair of treatment systems ○ Operational water quality testing OPS/FAC • Funding ○ Special water quality studies ○ Regional FAC Training • Certificate to Operate Program (CTO) ○ On-site sanitary surveys ○ CTO Report ○ Annual reviews

APPENDIX A: DEPARTMENT WATER QUALITY STANDARDS

Department water quality standards for water used for human consumption are based on the USEPA Primary and Secondary Drinking Water Standards¹³. USEPA Primary and Secondary Drinking Water Standards apply to water provided/distributed by Post (treated or untreated city water, well water, trucked water, or internally bottled and distributed water).

Department standards for commercially bottled water (when considered necessary) require that the bottled water be:

- Certified by the NSF International (NSF),
- From a reliable source (e.g. US military certified, or endorsed by the International Bottled Water Association (IBWA) or UL Certified.

The United States Food and Drug Administration (USFDA) has developed regulations and standards for bottled water which are used as the basis for certification by the IBWA and UL¹⁴, and are based on the USEPA Primary and Secondary Drinking Water Standards.

The USEPA, USFDA, and IBWA standards are listed in the tables below.

The USEPA standards have been established¹⁵ to protect public health by limiting the levels of specific contaminants in drinking water that can adversely affect public health and are known or anticipated to occur in water. The standards for primary drinking water contaminants are based on maximum contaminant level goals (MCLGs), maximum contaminant levels (MCLs), and treatment techniques (TT). MCLGs have been set for all primary drinking water contaminants - these are non-enforceable public health goals which are set at levels below which there is no known or expected risk to health. The MCLs are the highest level of a contaminant that is allowed in drinking water and are set as close to the MCLGs as feasible. ***The MCLs are the enforceable standards in the United States.*** In some cases, there is no reliable method to measure a contaminant at concentrations to indicate there is not a public health concern. For these contaminants, the USEPA sets a treatment techniques (TT) rather than an MCL. A TT describes a level of treatment needed to ensure control of a contaminant.

¹³ <https://www.epa.gov/ground-water-and-drinking-water/national-primary-drinking-water-regulations>

¹⁴ USFDA: The United States Food and Drug Administration, Code of Federal Regulations – Beverages and Bottled Water: Chapter I, Title 21, Part 165, Subpart B, Section 165.110.
<https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/cfrsearch.cfm?fr=165.110#:~:text=Sec.,safe%20and%20suitable%20antimicrobial%20agents>.

¹⁵ How EPA regulates drinking water contaminants: <https://www.epa.gov/sdwa/how-epa-regulates-drinking-water-contaminants>

Disinfectants (such as chlorine) are treated slightly differently. Maximum Residual Disinfectant Level Goal (MRDLG) identify the level of a drinking water disinfectant below which there is no known or expected risk to health. Like the MCLG, they are non-enforceable goals. The standards for disinfectants are based on the Maximum Residual Disinfectant Level (MRDL), which is the highest level of a disinfectant allowed in drinking water in the United States.

The US EPA Primary Drinking Water Standards, fall into five (5) categories of water contaminants that are of concern when assessing safe drinking water:

- Microorganisms;
- Disinfectants and disinfectant by-products;
- Inorganic chemicals;
- Organic chemicals; and
- Radionuclides.

USEPA Secondary Drinking Water¹⁶ are water quality standards for 15 contaminants that have aesthetic considerations, such as taste, color, and odor. These contaminants do not present a risk to human health at the secondary maximum contaminant levels (SMCLs).

¹⁶ <https://www.epa.gov/dwstandardsregulations/secondary-drinking-water-standards-guidance-nuisance-chemicals>

Tables of Primary Drinking Water Standards (Health Effects) and Bottled Water Standards

<u>Microorganisms</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBWA Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short- term)	Sources of Contaminant in Drinking Water		
<u>Total Coliforms</u> (including fecal coliform and <i>E. coli</i>)	zero	5.0% ⁴	Not a health threat in itself; it is used to indicate whether other potentially harmful bacteria may be present ⁵	Coliforms are naturally present in the environment; as well as feces; fecal coliforms and <i>E. coli</i> only come from human and animal fecal waste.	<u>Total Coliform</u> ⁴ MPN Method: No more than one bottle with > 2.2/100 mL AND No bottles with ≥ 9.2/100 milliliters No more than 1 bottle with 4.0 cfu/100 mL AND the arithmetic mean of the coliform density of the sample ≤ 1 cfu/100 mL <u>E.Coli</u> Zero	No <i>Escherichia coli</i> detectable in a 100 ml portion/sample. No validated total coliform detectable in a 100 mL portion/sample as substantiated by resampling. NOTE: Confirmation AND validation of all positive total coliform results in finished product required.

<u>Microorganisms</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBWA Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short- term)	Sources of Contaminant in Drinking Water		
<u>Heterotrophic plate count (HPC)</u>	n/a	TT ³	HPC has no health effects; it is an analytic method used to measure the variety of bacteria that are common in water. The lower the concentration of bacteria in drinking water, the better maintained the water system is.	HPC measures a range of bacteria that are naturally present in the environment	-	-
<u>Turbidity</u>	n/a	TT ³ ≤1 NTU in all samples	Turbidity is a measure of the cloudiness of water. It is used to indicate water quality and filtration effectiveness (such as	Soil runoff	≤5 NTU	≤0.5 NTU

<u>Microorganisms</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBWA Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short- term)	Sources of Contaminant in Drinking Water		
		AND ≤ 0.3 NTUs in at least 95 percent of the samples in any month.	whether disease- causing organisms are present). Higher turbidity levels are often associated with higher levels of disease-causing microorganisms such as viruses, parasites and some bacteria. These organisms can cause symptoms such as nausea, cramps, diarrhea, and associated headaches.			
<u>Legionella</u>	zero	TT ³	Legionnaire's Disease, a type of pneumonia	Found naturally in water; multiplies in heating systems	-	-

<u>Microorganisms</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBWA Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short- term)	Sources of Contaminant in Drinking Water		
<u>Cryptosporidium</u>	zero	TT ³	Gastrointestinal illness (such as diarrhea, vomiting, and cramps)	Human and animal fecal waste	-	-
<u>Giardia lamblia</u>	zero	TT ³	Gastrointestinal illness (such as diarrhea, vomiting, and cramps)	Human and animal fecal waste	-	-
<u>Viruses (enteric)</u>	zero	TT ³	Gastrointestinal illness (such as diarrhea, vomiting, and cramps)	Human and animal fecal waste	-	-

<u>Disinfectants and Disinfection Byproducts (USEPA Primary Drinking Water Quality Standards)</u>				
Contaminant	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water
<u>Disinfectants</u>				
<u>Chloramines (as Cl₂)</u>	MRDLG=4 ¹	MRDL=4.0 ¹	Eye/nose irritation; stomach discomfort, anemia	Water additive used to control microbes
<u>Chlorine (as Cl₂)</u>	MRDLG=4 ¹	MRDL=4.0 ¹	Eye/nose irritation; stomach discomfort	Water additive used to control microbes
<u>Chlorine dioxide (as ClO₂)</u>	MRDLG=0.8 ¹	MRDL=0.8 ¹	Anemia; infants and young children: nervous system effects	Water additive used to control microbes
<u>Disinfection By-Products</u>				
<u>Total Trihalomethanes (TTHMs)</u>	n/a ⁶	> 0.080	Liver, kidney or central nervous system problems; increased risk of cancer	Byproduct of drinking water disinfection
<u>Haloacetic acids (HAA5)</u>	n/a ⁶	0.060	Increased risk of cancer	Byproduct of drinking water disinfection

Disinfectants and Disinfection Byproducts (USEPA Primary Drinking Water Quality Standards)				
Contaminant	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water
<u>Bromate</u>	zero	0.010	Increased risk of cancer	Byproduct of drinking water disinfection
<u>Chlorite</u>	0.8	1.0	Anemia; infants and young children: nervous system effects	Byproduct of drinking water disinfection

<u>Inorganic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Antimony</u>	0.006	0.006	Increase in blood cholesterol; decrease in blood sugar	Discharge from petroleum refineries; fire retardants; ceramics; electronics; solder	0.006	0.006
<u>Arsenic</u>	0	0.010 as of 01/23/06	Skin damage or problems with circulatory systems, and may have increased risk of getting cancer	Erosion of natural deposits; runoff from orchards, runoff from glass and electronicsproduction wastes	0.010	0.01
<u>Asbestos (fiber ≥ 10 micrometers)</u>	7 million fibers per liter (MFL)	7 MFL	Increased risk of developing benign intestinal polyps	Decay of asbestos cement in water mains; erosion of natural deposits	-	-

Inorganic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Barium</u>	2	2	Increase in blood pressure	Discharge of drilling wastes; discharge from metal refineries; erosion of natural deposits	2	1
<u>Beryllium</u>	0.004	0.004	Intestinal lesions	Discharge from metal refineries and coal- burning factories; discharge from electrical, aerospace, and defense industries	0.004	0.004
<u>Cadmium</u>	0.005	0.005	Kidney damage	Corrosion of galvanized pipes; erosion of natural deposits; discharge from metal refineries; runoff from waste batteries and paints	0.005	0.005
<u>Chromium</u>	0.1	0.1	Allergic dermatitis	Discharge from steel and	0.1	0.05

<u>Inorganic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>(total)</u>				pulp mills; erosion of natural deposits		
<u>Copper</u>	1.3	TT ² ; Action Level=1.3	Short term exposure: Gastrointestinal distress Long term exposure: Liver or kidney damage People with Wilson's Disease should consult their personal doctor if the amount of copper in their water exceeds the action level	Corrosion of household plumbing systems; erosion of natural deposits	1.0	-
<u>Cyanide (as free cyanide)</u>	0.2	0.2	Nerve damage or thyroid problems	Discharge from steel/metal factories; discharge from plastic and fertilizer factories	0.2	0.2

Inorganic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Fluoride</u>	4.0	4.0	Bone disease (pain and tenderness of the bones); Children may get mottled teeth	Water additive which promotes strong teeth; erosion of natural deposits; discharge from fertilizer and aluminum factories	see USFDA regulation FR 165.110	SOQ dependent upon temperature and other factors. See IBWA Code of Practice.
<u>Lead</u>	zero	TT ² ; Action Level=0.015	Infants and children: Delays in physical or mental development; children could show slight deficits in attention span and learning abilities Adults: Kidney problems; high blood pressure	Corrosion of household plumbing systems; erosion of natural deposits	0.005	0.005
<u>Mercury</u> <u>(inorganic)</u>	0.002	0.002	Kidney damage	Erosion of natural deposits; discharge from refineries and factories; runoff from landfills and croplands	0.002	0.001

Inorganic Chemicals						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Nitrate</u> (measured as <u>Nitrogen</u>)	10	10	Infants below the age of six months who drink water containing nitrate in excess of the MCL could become seriously ill and, if untreated, may die. Symptoms include shortness of breath and blue-baby syndrome.	Runoff from fertilizer use; leaking from septic tanks, sewage; erosion of natural deposits	10	10
<u>Nitrite</u> (measured as <u>Nitrogen</u>)	1	1	Infants below the age of six months who drink water containing nitrite in excess of the MCL could become seriously ill and, if untreated, may die. Symptoms include shortness of breath and blue-baby syndrome.	Runoff from fertilizer use; leaking from septic tanks, sewage; erosion of natural deposits	1	1
Total Nitrate and Nitrite	-	-	-	-	10	10

Inorganic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards (mg/L)	IBWA Standard of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Selenium</u>	0.05	0.05	Hair or fingernail loss; numbness in fingers or toes; circulatory problems	Discharge from petroleum refineries; erosion of natural deposits; discharge from mines	0.05	0.01
<u>Thallium</u>	0.0005	0.002	Hair loss; changes in blood; kidney, intestine, or liver problems	Leaching from ore- processing sites; discharge from electronics, glass, and drug factories	0.002	0.002

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long- Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Acrylamide</u>	zero	TT ³	Nervous system or blood problems; increased risk of cancer	Added to water during sewage/wastewater treatment	-	
<u>Alachlor</u>	zero	0.002	Eye, liver, kidney or spleen problems; anemia; increased risk of cancer	Runoff from herbicide used on row crops	0.002	0.002
<u>Atrazine</u>	0.003	0.003	Cardiovascular system or reproductive problems	Runoff from herbicide used on row crops	0.003	0.003
<u>Benzene</u>	zero	0.005	Anemia; decrease in blood platelets; increased risk of cancer	Discharge from factories; leaching from gas storage tanks and landfills	0.005	

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Benzo(a)pyrene (PAHs)</u>	zero	0.0002	Reproductive difficulties; increased risk of cancer	Leaching from linings of water storage tanks and distribution lines	0.0002	0.0002
<u>Carbofuran</u>	0.04	0.04	Problems with blood, nervous system, or reproductive system	Leaching of soil fumigant used on rice and alfalfa	0.04	0.04
<u>Carbon tetrachloride</u>	zero	0.005	Liver problems; increased risk of cancer	Discharge from chemical plants and other industrial activities	0.005	0.002
<u>Chlordane</u>	zero	0.002	Liver or nervous system problems; increased risk of cancer	Residue of banned termiticide	0.002	0.00005
<u>Chlorobenzene</u>	0.1	0.1	Liver or kidney	Discharge from chemical	0.1	0.05

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
			problems	and agricultural chemical factories		
<u>2,4-D</u>	0.07	0.07	Kidney, liver, or adrenal gland problems	Runoff from herbicide used on row crops	0.07	0.07
<u>Dalapon</u>	0.2	0.2	Minor kidney changes	Runoff from herbicide used on rights of way	0.2	0.2
<u>1,2-Dibromo-3-chloropropane (DBCP)</u>	zero	0.0002	Reproductive difficulties; increased risk of cancer	Runoff/leaching from soil fumigant used on soybeans, cotton, pineapples, and orchards	0.0002	0.0002
<u>o-Dichlorobenzene</u>	0.6	0.6	Liver, kidney, or circulatory system problems	Discharge from industrial chemical factories	0.6	0.6

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>p-Dichlorobenzene</u>	0.075	0.075	Anemia; liver, kidney or spleen damage; changes in blood	Discharge from industrial chemical factories	0.075	0.075
<u>1,2-Dichloroethane</u>	zero	0.005	Increased risk of cancer	Discharge from industrial chemical factories	0.005	0.002
<u>1,1-Dichloroethylene</u>	0.007	0.007	Liver problems	Discharge from industrial chemical factories	0.007	0.002
<u>cis-1,2-Dichloroethylene</u>	0.07	0.07	Liver problems	Discharge from industrial chemical factories	0.07	0.07
<u>trans-1,2-Dichloroethylene</u>	0.1	0.1	Liver problems	Discharge from industrial chemical factories	0.1	0.1
<u>Dichloromethane</u>	zero	0.005	Liver problems;	Discharge from drug and	0.005	0.003

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
			increased risk of cancer	chemical factories		
<u>1,2-Dichloropropane</u>	zero	0.005	Increased risk of cancer	Discharge from industrial chemical factories	0.005	0.005
<u>Di(2-ethylhexyl) adipate</u>	0.4	0.4	Weight loss, liver problems, or possible reproductive difficulties.	Discharge from chemical factories	0.4	0.4
<u>Di(2-ethylhexyl) phthalate</u>	zero	0.006	Reproductive difficulties; liver problems; increased risk of cancer	Discharge from rubber and chemical factories	0.006	0.006
<u>Dinoseb</u>	0.007	0.007	Reproductive difficulties	Runoff from herbicide used on soybeans and vegetables	0.007	0.007

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Dioxin (2,3,7,8-TCDD)</u>	zero	0.00000003	Reproductive difficulties; increased risk of cancer	Emissions from waste incineration and other combustion; discharge from chemical factories	0.00000003	0.00000003
<u>Diquat</u>	0.02	0.02	Cataracts	Runoff from herbicide use	0.02	0.02
<u>Endothall</u>	0.1	0.1	Stomach and intestinal problems	Runoff from herbicide use	0.1	0.1
<u>Endrin</u>	0.002	0.002	Liver problems	Residue of banned insecticide	0.002	0.002
<u>Epichlorohydrin</u>	zero	TT ³	Increased cancer risk, and over a long period of time, stomach	Discharge from industrial chemical factories; an impurity of some water	-	-

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
			problems	treatment chemicals		
<u>Ethylbenzene</u>	0.7	0.7	Liver or kidneys problems	Discharge from petroleum refineries	0.7	0.7
<u>Ethylene dibromide</u>	zero	0.00005	Problems with liver, stomach, reproductive system, or kidneys; increased risk of cancer	Discharge from petroleum refineries	0.00005	0.00005
<u>Glyphosate</u>	0.7	0.7	Kidney problems; reproductive difficulties	Runoff from herbicide use	0.7	0.7
<u>Heptachlor</u>	zero	0.0004	Liver damage; increased risk of cancer	Residue of banned termiticide	0.0004	0.0004

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Heptachlor epoxide</u>	zero	0.0002	Liver damage; increased risk of cancer	Breakdown of heptachlor	0.0002	0.0002
<u>Hexachlorobenzene</u>	zero	0.001	Liver or kidney problems; reproductive difficulties; increased risk of cancer	Discharge from metal refineries and agricultural chemical factories	0.001	0.001
<u>Hexachlorocyclopentadiene</u>	0.05	0.05	Kidney or stomach problems	Discharge from chemical factories	0.05	0.05
<u>Lindane</u>	0.0002	0.0002	Liver or kidney problems	Runoff/leaching from insecticide used on cattle, lumber, gardens	0.0002	0.0002
<u>Methoxychlor</u>	0.04	0.04	Reproductive difficulties	Runoff/leaching from insecticide used on fruits,	0.04	0.04

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
				vegetables, alfalfa, livestock		
Monochlorobenzene (108-90-7)	-	-	-	-	0.1	0.05
<u>Oxamyl (Vydate)</u>	0.2	0.2	Slight nervous system effects	Runoff/leaching from insecticide used on apples, potatoes, and tomatoes	0.2	0.2
<u>Polychlorinated biphenyls (PCBs)</u>	zero	0.0005	Skin changes; thymus gland problems; immune deficiencies; reproductive or nervous system difficulties; increased risk of cancer	Runoff from landfills; discharge of waste chemicals	0.0005	0.0005

Organic Chemicals

Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
<u>Pentachlorophenol</u>	zero	0.001	Liver or kidney problems; increased cancer risk	Discharge from wood preserving factories	-	0.001
<u>Picloram</u>	0.5	0.5	Liver problems	Herbicide runoff	0.5	0.5
<u>Simazine</u>	0.004	0.004	Problems with blood	Herbicide runoff	0.0004	0.004
<u>Styrene</u>	0.1	0.1	Liver, kidney, or circulatory system problems	Discharge from rubber and plastic factories; leaching from landfills	0.1	0.1
<u>Tetrachloroethylene</u>	zero	0.005	Liver problems; increased risk of cancer	Discharge from factories and dry cleaners	0.005	0.001
<u>Toluene</u>	1	1	Nervous system, kidney,	Discharge from	1	1

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
			or liver problems	petroleum factories		
<u>Toxaphene</u>	zero	0.003	Kidney, liver, or thyroid problems; increased risk of cancer	Runoff/leaching from insecticide used on cotton and cattle	0.003	0.003
<u>2,4,5-TP (Silvex)</u>	0.05	0.05	Liver problems	Residue of banned herbicide	0.05	0.01
<u>1,2,4-Trichlorobenzene</u>	0.07	0.07	Changes in adrenal glands	Discharge from textile finishing factories	0.07	0.009
<u>1,1,1-Trichloroethane</u>	0.20	0.2	Liver, nervous system, or circulatory problems	Discharge from metal degreasing sites and other factories	0.2	0.03
<u>1,1,2-Trichloroethane</u>	0.003	0.005	Liver, kidney, or	Discharge from industrial	0.005	0.003

<u>Organic Chemicals</u>						
Contaminant	USEPA Primary Drinking Water Quality Standards				USFDA Bottled Water Quality Standards	IBW Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water		
			immune system problems	chemical factories		
<u>Trichloroethylene</u>	zero	0.005	Liver problems; increased risk of cancer	Discharge from metal degreasing sites and other factories	0.005	0.001
<u>Vinyl chloride</u>	zero	0.002	Increased risk of cancer	Leaching from PVC pipes; discharge from plastic factories	0.002	0.002
<u>Xylenes (total)</u>	10	10	Nervous system damage	Discharge from petroleum factories; discharge from chemical factories	10	1

<u>Radionuclides</u>					
Contaminant	USEPA Primary Drinking Water Quality Standards				IBWA Standards of Quality (SOQ)
	MCLG ¹ (mg/L) ²	MCL or TT ¹ (mg/L) ²	Potential Health Effects from Long-Term Exposure Above the MCL (unless specified as short-term)	Sources of Contaminant in Drinking Water	
<u>Alpha particles</u>	zero	15 picocuries per Liter (pCi/L)	Increased risk of cancer	Erosion of natural deposits of certain minerals that are radioactive and may emit a form of radiation known as alpha radiation	15 picocuries per Liter (pCi/L)
<u>Beta particles and photon emitters</u>	zero	4 millirems per year	Increased risk of cancer	Decay of natural and man-made deposits of certain minerals that are radioactive and may emit forms of radiation known as photons and beta radiation	50 pCi/L
<u>Radium 226 and Radium 228 (combined)</u>	zero	5 pCi/L	Increased risk of cancer	Erosion of natural deposits	5 pCi/L
<u>Uranium</u>	zero	0.030 as of 12/08/03	Increased risk of cancer, kidney toxicity	Erosion of natural deposits	0.030

Table of Secondary Drinking Water Standards (Non-Health/Aesthetic Effects)

Contaminant	USEPA Secondary Water Quality Standards		USFDA Bottled Water Quality Standards**	IBWQ Standards of Quality (SOQ)
	Secondary MCL	Noticeable Effects above the Secondary MCL		
Aluminum	0.05 to 0.2 mg/L*	colored water		0.2
Chloride	250 mg/L	salty taste	250 mg/L	250 mg/L
Color	15 color units	visible tint	15 color units	5 color units
Copper	1.0 mg/L	metallic taste; blue-green staining		1.0 mg/L
Corrosivity	Non-corrosive	metallic taste; corroded pipes/ fixtures staining	-	-
Fluoride	2.0 mg/L	tooth discoloration	-	-
Foaming agents	0.5 mg/L	frothy, cloudy; bitter taste; odor	-	-
Iron	0.3 mg/L	rusty color; sediment; metallic taste; reddish or orange staining	0.3 mg/L	0.3 mg/L

Contaminant	USEPA Secondary Water Quality Standards		USFDA Bottled Water Quality Standards**	IBWQ Standards of Quality (SOQ)
	Secondary MCL	Noticeable Effects above the Secondary MCL		
Manganese	0.05 mg/L	black to brown color; black staining; bitter metallic taste	0.05 mg/L	0.05 mg/L
Odor	3 TON (threshold odor number)	“rotten-egg”, musty or chemical smell	3 TON (threshold odor number)	3 TON (threshold odor number)
pH	6.5 – 8.5	low pH: bitter metallic taste; corrosion high pH: slippery feel; soda taste; deposits		5 – 7 (purified water) 6.5 – 8.5 (source water and other product waters)
Silver	0.1 mg/L	skin discoloration; graying of the white part of the eye	0.1 mg/L	0.025 mg/L
Sulfate	250 mg/L	salty taste	250 mg/L	250 mg/L
Total Dissolved Solids (TDS)	500 mg/L	hardness; deposits; colored water; staining; salty taste	500 mg/L	500 mg/L

Contaminant	USEPA Secondary Water Quality Standards		USFDA Bottled Water Quality Standards**	IBWQ Standards of Quality (SOQ)
	Secondary MCL	Noticeable Effects above the Secondary MCL		
Zinc	5 mg/L	metallic taste	5 mg/L	5 mg/L
Phenols	-	-	0.05 mg/L	0.001 mg/L

*mg/L is milligrams of substance per liter of water.

**Mineral water is except from allowable level. The exceptions are aesthetically based allowable levels and do not relate to a health concern.

Additional International Bottled Water Association (IBWA) Standards of Quality (SOQ)

Contaminant	IBWA SOQ (mg/L)
Methyl tertiary butyl ether (MTBE)	0.07
Napthalene	0.3
Perchlorate	0.002
m-Dichlorobenzene	0.6
1,1-Dichloroethane	0.05
1,1,2,2-Tetrachloroethane	0.001
Per- and Polyfluoroalkyl Substances (PFAS) (US EPA Method 537.1)	5 ng/L (single detect) 10 ng/L (multiple detect)

Notes on USEPA Primary Drinking Water Standards

¹Definitions:

Maximum Contaminant Level Goal (MCLG) - The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety and are non-enforceable public health goals.

Maximum Contaminant Level (MCL) - The highest level of a contaminant that is allowed in drinking water. MCLs are set as close to MCLGs as feasible using the best available treatment technology and taking cost into consideration. MCLs are enforceable standards.

Maximum Residual Disinfectant Level Goal (MRDLG) - The level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contaminants.

Treatment Technique (TT) - A required process intended to reduce the level of a contaminant in drinking water.

Maximum Residual Disinfectant Level (MRDL) - The highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

² Units are in milligrams per liter (mg/L) unless otherwise noted. Milligrams per liter are equivalent to parts per million (PPM).

³ EPA's surface water treatment rules require systems using surface water or ground water under the direct influence of surface water to 1) Disinfect their water, and 2) Filter their water, or

Meet criteria for avoiding filtration so that the following contaminants are controlled at the following levels:

Cryptosporidium: Unfiltered systems are required to include *Cryptosporidium* in their existing watershed control provisions

Giardia lamblia: 99.9% removal/inactivation.

Viruses: 99.99% removal/inactivation.

Legionella: No limit, but EPA believes that if *Giardia* and viruses are removed/inactivated, according to the treatment techniques in the Surface Water Treatment Rule, *Legionella* will also be controlled.

Turbidity: For systems that use conventional or direct filtration, at no time can turbidity (cloudiness of water) go higher than 1 Nephelometric Turbidity Unit (NTU), and samples for turbidity must be less than or equal to 0.3 NTUs in at least 95 percent of the samples in any month. Systems that use filtration other than the conventional or direct filtration must follow state limits, which must include turbidity at no time exceeding 5 NTUs.

Heterotrophic Plate Count (HPC): No more than 500 bacterial colonies per milliliter. (Does not typically apply to DoS Potable Water Treatment systems).

Long Term 1 Enhanced Surface Water Treatment: Surface water systems or groundwater under the direct influence (GWUDI) systems serving fewer than 10,000 people must comply with the applicable Long Term 1 Enhanced Surface Water Treatment Rule provisions (such as turbidity standards, individual filter monitoring, *Cryptosporidium* removal requirements, updated watershed control requirements for unfiltered systems). (Does not typically apply to DoS Potable Water Treatment systems).

Long Term 2 Enhanced Surface Water Treatment Rule: This rule applies to all surface water systems or ground water systems under the direct influence of surface water. The rule targets additional *Cryptosporidium* treatment requirements for higher risk systems and includes provisions to reduce risks from uncovered finished water storage facilities and to ensure that the systems maintain microbial protection as they take steps to reduce the formation of disinfection byproducts. (Does not typically apply to DoS Potable Water Treatment systems).

Filter Backwash Recycling: This rule requires systems that recycle to return specific recycle flows through all processes of the system's existing conventional or direct filtration system or at an alternate approved location

⁴ USEPA Primary Drinking Water Quality Standards: No more than 5.0% samples total coliform-positive (TC-positive) in a month. (For water systems that collect fewer than 40 routine samples per month, no more than one sample can be total coliform-positive per month.) Every sample that has total coliform must be analyzed for either fecal coliforms or *E. coli*. If two consecutive TC-positive samples, and one is also positive for *E.coli* fecal coliforms, system has an acute MCL violaton.

⁴USFDA Bottled Water Quality Standards: MPN Method - Not more than one of the analytical units in the sample shall have a most probable number (MPN) of 2.2 or more coliform organisms per 100 milliliters and no analytical unit shall have an MPN of 9.2 or more coliform organisms per 100 milliliters. MF Method - Not more than one of the analytical units in the sample shall have 4.0 or more coliform organisms per 100 milliliters and the arithmetic mean of the coliform density of the sample shall not exceed one coliform organism per 100 milliliters

⁵ Fecal coliform and *E. coli* are bacteria whose presence indicates that the water may be contaminated with human or animal wastes. Disease-causing microbes (pathogens) in these wastes can cause diarrhea, cramps, nausea, headaches, or other symptoms. These pathogens may pose a special health risk for infants, young children, and people with severely compromised immune systems.

⁶ Although there is no collective MCLG for this contaminant group, there are individual MCLGs for some of the individual contaminants:

Trihalomethanes: bromodichloromethane (zero); bromoform (zero); dibromochloromethane (0.06 mg/L); chloroform (0.07 mg/L).

Haloacetic acids: dichloroacetic acid (zero); trichloroacetic acid (0.02 mg/L); monochloroacetic acid (0.07mg/L). Bromoacetic acid and dibromoacetic acid are regulated with this group but have no MCLGs.

⁷ Lead and copper are regulated by a treatment technique that requires systems to control the corrosiveness of their water. If more than 10% of **tap water samples** exceed the action level, water systems must take additional steps. For copper, the action level is 1.3 mg/L, and for lead is 0.015 mg/L.

⁸ (Does not typically apply to DoS Potable Water Treatment systems). Each water system must certify, in writing, (using third-party or manufacturer's certification) that when acrylamide and epichlorohydrin are used to treat water, the combination (or product) of dose and monomer level does not exceed the levels specified, as follows:

Acrylamide = 0.05% dosed at 1 mg/L (or equivalent)

Epichlorohydrin = 0.01% dosed at 20 mg/L (or equivalent)

APPENDIX B: MED FLUORIDE SUPPLEMENTATION AND FLUORIDE-CONTAINING TOOTHPASTE USE IN CHILDREN¹⁷

Evidence for caries prevention and fluorosis risk with **oral fluoride supplements** is incomplete. Published guidelines disagree regarding which children aged 6 months to 16 years of age should receive oral fluoride supplementation. Among the American Dental Association (ADA), American Academy of Pediatrics (AAP), Centers for Disease Control (CDC), United States Preventive Services Task Force (USPSTF), American Academy of Pediatric Dentistry (AAPD), and American Academy of Family Physicians (AAFP) there is some disagreement over the use of supplemental fluoride in children to prevent dental caries. Some of these groups recommend supplementing **all** children with inadequate fluoride content in their drinking water, while others advocate supplementation only children with inadequate fluoride content in their drinking water who are at **high risk** for caries.

Considerations primarily balance concerns for caries versus [fluorosis](#), a cosmetic problem. There is no validated tool for determining which children are at high risk of caries, but known risk factors to evaluate include:

- Previous caries
- White spot lesions or enamel defects
- Plaque accumulation
- Nighttime bottle use (other than water)
- Lower socioeconomic status
- Parental/ sibling history of caries and abscesses
- Lack of exposure to fluoride in water, toothpaste, and professional topical treatments
- Special health care needs (e.g. developmental disability, cleft lip/palate, low birth weight)

MED advises providers to engage parents in a discussion regarding benefits and risks of fluoride supplements and offer prescriptions to children ages 6 months to 16 years with inadequate dietary intake if parents or providers have concerns of caries risk.

If supplements are prescribed, base daily fluoride dose on age and water fluoride content¹⁸ per the ADA guidelines:

¹⁷ June 2022

¹⁸ Fluoride content of bottled water is variable, while fluoride content of distilled water is negligible. If you use a particular brand of bottled water consistently, the manufacturer may be able to supply that information. Fluoride content of Post provided/distributed water (i.e. treated or untreated city water, well water, trucked water, or internally bottled water), can be obtained from FAC Water Testing Group and/or SHEM.

Age	Water Fluoride Content	
	< 0.3 mg/L	0.3-0.6 mg/L
6 months – 3 years	0.25 mg	
3 years – 6 years	0.5 mg	0.25 mg
6 years – 16 years	1 mg	0.5 mg

Topical fluoride application via toothpaste offers additional caries prevention (at additional risk of fluorosis), but evidence for specific recommendations is also limited. AAP, CDC, and AAPD agree that children aged 2-6 years should use a “pea-sized” amount of fluoride-containing toothpaste twice a day. ADA limits that age range to 3-7 years.

ADA and AAPD recommend starting a “smear” or “rice-sized” amount of fluoride-containing toothpaste at first tooth eruption, while AAP and CDC recommend discussing risks and benefits with parents before starting in children younger than 2 years.

MED advises discussing with parents whether to start a “smear” of fluoride-containing toothpaste prior to 2 years of age. We advise recommending that children use a “pea-sized” amount of toothpaste twice a day from age 2 through 6 years. Children should be encouraged to spit excess toothpaste.

Web references

- AAPD: http://www.aapd.org/media/Policies_Guidelines/P_FluorideUse.pdf
www.aapd.org/media/Policies_Guidelines/G_CariesRiskAssessment.pdf
AAP: http://www2.aap.org/ORALHEALTH/pact/ch6_sect3b.cfm
ADA: http://ebd.ada.org/~media/EBD/Files/ADA_Evidence-based_Fluoride_Supplement_Chairside%20Guide.ashx
CDC: <http://www.cdc.gov/mmwr/preview/mmwrhtml/rr5014a1.htm>
USPSTF: <http://www.uspreventiveservicestaskforce.org/Page/Document/UpdateSummaryFinal/dental-caries-in-children-from-birth-through-age-5-years-screening>
AAFP: <http://www.aafp.org/patient-care/clinical-recommendations/all/dental-caries.html>

APPENDIX C: DRINKING WATER SAFETY PROGRAM DOCUMENT - EXAMPLES

Example Table of Contents

1. Identify the Water Safety Team (Roles and Responsibilities)
 - 1.1. Management
 - 1.2. Facilities
 - 1.3. POSHO
 - 1.4. Health Unit
2. System Description
 - 2.1. Source Water
 - 2.1.1. Type (surface, ground, trucked, bottled)
 - 2.1.2. Provider (local utility, trucked water company, commercially bottled)
 - 2.1.3. Source water quality assessment (surface water, ground water, trucked water, bottled water) (with SHEM)
 - 2.1.4. Justification for Commercially Bottled Water (if applicable, with SHEM)
 - 2.2. Treatment System (describe storage, treatment, distribution) as applicable
3. Operations and Maintenance Procedures
 - 3.1. Routine O&M
 - 3.1.1. Roles and Responsibilities
 - 3.1.1.1. Maintenance checklists and work orders (GMMS assets for O&M)
 - 3.1.1.2. Monitoring
 - 3.1.1.2.1. OpMon/QC Test Plan (collection, analysis, reporting)
 - 3.1.1.2.2. Communications (FAC, HU, Management)
 - 3.1.1.3. Response procedures
 - 3.1.1.3.1. Actions
 - 3.1.1.3.2. Communications
 - 3.2. Incidents (describe procedures, communications, follow-up for the following:)
 - 3.2.1. Microbiological incidents
 - 3.2.2. Chemical incidents
 - 3.2.3. Contamination
 - 3.2.3.1. Source contamination
 - 3.2.3.1.1. Fuel Spill
 - 3.2.3.1.2. Other
 - 3.2.3.2. Distribution contamination
 - 3.2.3.2.1. Pipe breaks
 - 3.2.3.2.2. Backflow incident
 - 3.3. Certificate to Operate Program (describe status if applicable)
 4. Verification of Quality
 - 4.1. Roles and Responsibilities
 - 4.1.1. QC Testing (collection, analysis, reporting)
 - 4.1.2. Potability determinations (coordination with SHEM)

4.1.3. Communications at Post

5. Updating the DWSP

5.1. Schedule

5.2. Roles and Responsibilities



No XX/2023

Month/Day/Year

To: Facilities and Health Unit Staff and Supervisors

Subject: Drinking Water Safety Program at <<Name Post>>

References: [15 FAM 957.5](#)

Post must provide potable water that is microbiologically and chemically safe to drink at all Department of State-owned or -leased facilities abroad where employees and post family members work or reside.

This program outlines a process to identify and control water contaminants of greatest health concern and provide assurance that reliably safe drinking water is available at post.

Responsibilities

Post Occupational Safety and Health Officer (POSHO) is responsible that requirements related to selection, operation, and maintenance of point-of-use (POU) and point-of-entry (POE) devices and equipment are met.

POSHO Assistant (APOSHO) will assist POSHO as necessary.

FAC Residential and NEC Supervisors and Plumbers are responsible for collecting water samples per the Operational Monitoring/QC Testing Plan (Appendix B) and maintaining system and devices in good and operable condition. Maintenance schedule for drinking water treatment systems is available in GMMS.

FAC NEC Supervisor and Plumbers are responsible for collecting and testing water samples per the Operational Monitoring/QC Testing Plan (Appendix B), with the exception of Total Coliforms and E.Coli samples which are analyzed by the Health Unit.

Health Unit is responsible for analyzing water samples for Total Coliforms and E. Coli and assisting POSHO in managing this program according to their best knowledge and capability. HU is also responsible for monitoring diseases that might be related to drinking water contaminants, and for sharing concerns and issues with POSHO and/or POSHO Assistant.

Raw Water Quality Assessment

Post must review all available raw water quality information with the assistance of SHEM. Based on it, post must assess the water quality based on available information and identify contaminants of concern. The goal of evaluating the raw water is to determine if it is potable and, if it is not potable, to identify what types of contaminants must be removed to render it

potable.

Water Treatment Options

Post employs two categories of water treatment systems. Point-of-entry (POE) systems are the larger systems that may treat water for an entire building or compound. Point-of-use (POU) devices are those employed at individual cold water taps to provide potable drinking water.

Maintenance, Repair and Cleaning of Water Storage Systems

All components of the water treatment, storage, and distribution system within the control of post must be maintained to foster good water quality.

Post must institute a systematic, routine maintenance/preventive maintenance program in accordance with instructions of the POE and POU manufacturers.

Potable water storage tank cleaning must be done on an as needed basis as determined by conducting periodic visual tank inspections done from outside the tank.

Quality Control Monitoring

Post must monitor the microbiological quality of drinking water. Sampling must be conducted at 10% of drinking water taps in buildings/residences. The Operational Monitoring/Quality Control Testing Plan identifies the schedule (Appendix B) and the testing that is completed. It may be changed independently from this document if necessary (new residences leased, old residences decommissioned, etc.). Actual schedule is updated by APOSHO in a separate file.

Sample Collection Procedure

Special care is needed during microbiological sampling to ensure that contamination does not occur as a result of the sampling process. At post, FAC Residential Plumbers sample water from residences, FAC NEC Plumbers sample water from NEC buildings. Plumbers must follow the procedure below:

1. Wash your hands thoroughly with soap and water or use a disinfecting wipe prior to beginning the sampling procedure.
2. Remove any attachment from the faucet, including aerators, screens, washers, hoses, and water filters. Disinfect the sample site before collecting the samples with an alcohol swab or bleach, be sure to flush the site thoroughly to remove all disinfectant.
3. Turn on the cold water only and let it run with a steady stream for at least 3 (three) minutes. Before collecting the sample, turn the water down to a thin stream (about the width of a pencil), then let the water run one minute. If the system is chlorinated, measure the free chlorine residual, and note the measurement.
4. To avoid contamination while taking the sample, hold the bottle near the bottom with one hand, hold the top of the cap with the other, and then unscrew the cap.

Note: Do not rinse out powder to neutralize chlorine in the sample bottle.

Note: Do not set the cap down, touch any part of the cap that touches the bottle, or let anything touch the rim of the bottle or the inside of the cap.

5. Hold the bottle under the stream of water. Be careful not to let the bottle touch the sample tap. Fully fill the bottle to the neck or indicated fill line. Don't allow it to overflow. Remove the bottle from the water flow and carefully screw the cap back on.

Resampling and Retesting

Because sample contamination is common, all positive Total Coliform tests should be retested

as soon as possible at the same tap, to confirm the presence of E. Coli bacteria. Follow the same steps as in Sample Collection Procedure.

Post must promptly respond to unacceptable quality monitoring results as described in the table below.

Finding	Response
Total Coliform- negative and E. Coli- negative	This indicates that the original positive result might be a false positive. Post should continue normal monitoring procedures.
Total Coliform- positive and E. Coli- negative	Coliform contamination confirmed but there are no fecal coliform indicators present. This does not require a boil order and residents do not have to stop consuming the water, but Post should contact Water Testing Group and SHEM, and begin troubleshooting the source of the contamination.
Total Coliform- positive and E. Coli- positive	Fecal coliform contamination is confirmed. Post should issue a boil order to affected personnel (see template) and contact Water Testing Group and SHEM. Post should begin troubleshooting the source of the contamination and follow the process outlined in this document for confirmatory testing to put the water system back into service.

Post should report all confirmed positive coliform tests to the POSHO and notify Water Testing Group and SHEM via email. When reporting results, please include detailed information about the test results and sampling location.

After the source of contamination has been identified and fixed, collect three samples on three separate days at the same location where the initial sample was confirmed positive for coliform.

Issuing Boil Order

If a retest sample confirms the presence of E. Coli/ Fecal Coliform, Post can no longer consider the water potable. Post should issue a Boil Order Notification to affected residents or personnel. Boil Order Notifications should be communicated to post management, Water Testing Group, SHEM and the Health Unit as well.

After the source of the contamination has been identified and fixed, collect three samples on three separate days at the same location where the initial sample was confirmed positive for coliform. The boil order should remain in effect until all 3 samples are negative for Total Coliform, indicating that the water is free of contamination.

Recording Test Results

POSHO Assistant (for residences) and NEC Plumbers (for NEC) must enter the results of all water quality tests and retests in FACApps, regardless of results. Use the link below to access the FACApps Water Quality Testing module where you can enter the test results:

https://facapps-usdos.msappproxy.net/cfsm_fac/esps/dev/index.cfm

Note: The Water Quality Testing Module is a user authenticated module and requires the user to have been granted prior access. If you do not have access, click REQUEST ACCESS button on the FACApps homepage.

1. Click the READING tab at the top of the screen.
2. Click SUBMIT READINGS.
3. Fill in the appropriate fields and click the SAVE and CONTINUE button to move to subsequent pages.

Note: Please ensure that you add the RPA Prop. ID to the Building field and that the Location field contains sufficient information to identify where the sample was taken (source water, water tanks, pre-treatment, post-treatment, etc.).

Bottled Water

If Post uses commercial bottled water, it must be certified or have quality hallmarks by the NSF, the US military (DoD), UL listing or endorsement by the International Bottled Water Association (IBWA). Otherwise, it is prohibited to use commercial bottled water. If the water has these quality hallmarks, quality control testing is not required.

To bottle water from a Point of Entry or Point of Use system, it is critical that stringent sanitary processes are followed to ensure water is not contaminated during the filling process which is, unfortunately, a common occurrence. Therefore, post prohibits bottling water for common use. However, post employees can bottle water for individual use.

APPENDIX D: PROCEDURES FOR SOURCE WATER QUALITY AND POTABILITY ASSESSMENTS

The processes for assessing source water quality and determining potability of water used for human consumption are detailed below.

Source Water Quality Assessments

City or Local Water Utility Source

OBO Responsibilities (OPS/SHEM). SHEM has a lead role in coordinating the assessment, researching host country practices, and reviewing water quality data. Specific activities that SHEM conducts during the source water evaluation are listed below:

- **Review of previous water source assessments.** SHEM reviews any previous water assessments and supporting documentation, including water quality testing data.
- **Review of legislative/regulatory framework.** SHEM reviews the host country's legislative and regulatory framework for water sanitation to determine what regulatory standards exist that control water treatment and distribution and to determine the variances with the US EPA Safe Drinking Water Standard (SDWA) requirements.
- **Review of public health infrastructure.** SHEM reviews the host country's public infrastructure to determine the effectiveness of health and disease tracking, public health notifications, and public health transparency.
- **Review of utility performance.** SHEM reviews any publicly available data about the municipal water utility performance, if applicable. Important information includes treatment processes and methods, financial information, non-revenue water rates, distribution system conditions and maintenance, and public transparency.
- **Review of water sources for municipal utility.** SHEM researches the municipal utility's water sources for security, contamination threats, and water quality. For groundwater sources that do not involve municipal treatment, SHEM reviews well-head security and water quality.
- **Coordination for water quality testing for initial and ongoing assessments.** Source water assessments require water testing. Testing for the EPA Safe Drinking Water Act (SDWA) Primary and Secondary regulated contaminants requires a certified lab capable of testing to EPA methods. SHEM assists with locating acceptable laboratories and coordinating water sampling and shipment logistics.
- **Potability determination and issuance of potability documentation.** As a result of the processes listed above, SHEM determines the potability classification of the source water and issues a memo documenting the potability status. The memo will document whether the water is fit for human consumption or whether it requires supplemental treatment for either contact use or consumption. SHEM will also

record the potability determination along with any research documentation in the Department of State Source Water Potability Assessment database.

Post Responsibility

- **Contact with local utility.** Post should identify in writing the name and contact information of a contact at the water utility who can provide information about the utility operations.
- **Organize ICASS funding for ongoing source water quality testing.** Water testing, both initial source water assessments and ongoing water testing, is an ICASS expense. Post is responsible for budgeting for and allocating ICASS funds for water testing.
- **Conducting water quality testing in coordination with SHEM.** Post conducts the water testing in coordination with SHEM. Depending upon the type of testing and facilities available at post and in the host country, this may include contracting with labs, water sampling, water testing, and arranging shipping logistics.
- **Reporting water quality testing results.** Water testing results should be reported to the FAC Water Testing Group and SHEM for evaluation. Post also enters the water testing data in the FACApps water testing module.
- **Reporting concerns about source water quality issues** and changes to water sources to SHEM and the FAC Water Testing Group.

Post Managed Groundwater Sources

OBO Responsibilities. OBO/OPS/SHEM has a lead role in coordinating the assessment and reviewing water quality data for potability. Specific activities that SHEM conducts during the source water evaluation are listed below:

- **Review of water quality data.** SHEM reviews any previous and current water quality testing data.
- **Review of well site and potential for contamination.** SHEM reviews the location and potential sources of contamination for the well, based on the Department's codes and standards for drilling wells.
- **Potability Determination.** SHEM renders a potability determination, and if the groundwater source doesn't meet Department's standards, and coordinates with OBO/FAC/PS Water Testing Group to identify appropriate treatment.

Post Responsibility

- **Conducting water quality testing.** For new groundwater sources, Post contracts with the well driller or conducts their own testing of the groundwater source per Department codes, standards, and specifications (contact FAC Water Testing Group for guidance on parameters to be tested on new wells). Depending upon the type of testing and facilities available at post and in the host country, this may include

contracting with labs, water sampling, water testing, and arranging shipping logistics.

- **Reporting water quality testing results.** Water testing results should be reported to the FAC Water Testing Group and SHEM for analysis. Post also enters the water testing data in the FACApps water testing module.
- **Reporting concerns about source water quality issues** and changes to water sources to SHEM and the FAC Water Testing Group

Commercially Bottled Water Sources

OBO Responsibilities. SHEM assesses the potability of commercially bottled water based on whether the supplier is certified or non-certified.

Certified Commercially Bottled Water. SHEM will declare commercially bottled water that is certified by: 1) the NSF, UL, or the US military (DoD), 2) endorsement by the International Bottled Water Association (IBWA), or 3) an acceptable Host Nation program used to certify bottled water as approved by SHEM (i.e. such as a Host Nation HAACP¹⁹ program). Post should provide verification by using the links below for certification and listing databases, and provide this verification to SHEM. If the commercially bottled water has these quality hallmarks, SHEM will approve them as potable sources. **Posts should obtain a copy of water quality testing results from the certified bottled water provider prior to contract award and annually during the life of the contract.** Post does not need to conduct quality control (QC) testing of certified commercially bottled water.

- **International Bottled Water Association (IBWA):**
<https://bottledwater.org/about-ibwa/>
- **Underwriters Laboratory Product IQ Database:**
<https://productiq.ulprospector.com/en/search>
- **Department of Defense Worldwide Directory of Sanitarily Approved Food Establishments:**
<https://phc.amedd.army.mil/topics/foodwater/ca/Pages/DoDApprovedFoodSources.aspx>
- **NSF International (NSF) Database:** <https://info.nsf.org/Certified/BWPI/>

Non-certified Commercially Bottled Water. Post Health Units have tested locally bottled water at overseas locations and reported a high incidence of total coliform and bacterial contaminants. In other cases, locally bottled water may be nothing more

¹⁹ HAACP = hazard analysis and critical control points. HAACP programs for certification of bottled water use a systematic approach to determine safety of bottled water from biological, chemical and physical hazards that may occur during production and transportation. The IBWA has adopted the HAACP system into its Model Code for certification of bottled water.

than bottled tap water. Because of this, if the quality of commercially bottled water cannot be verified through certification, SHEM will evaluate potability of non-certified commercially bottled water based on laboratory testing results that should be provided by the commercial supplier, and well as ongoing quality control (QC) testing completed by Post. The supplier should provide water quality test results indicating acceptable bottled water quality prior to contract award, as well as annually during the life of their contract. In addition, periodic quality control (QC) testing is required. Post should test a percentage of bottles received each month for quality control.

Summary of Commercially Bottled Water Assessments. Table D-1 lists the testing requirements for commercially bottled water (certified and non-certified). All water quality analyses completed by an outside lab should be done using an ISO 17025 certified laboratory.

Table D-1. Commercially Bottled Water Testing Requirements

Type of Commercially Bottled Water	Required Before Contract Award and Annually during Contract (Supplier responsibility)	QC Testing (Post responsibility)	SHEM Potability Determination
Certified	Proof of Certification and Water quality results (used to determine certification)	None	Based on initial Proof of Certification then annually
Non-Certified	Testing results for: • Total Coliform/e.Coli • Turbidity • TDS • pH • Temperature • Chlorine Residual (Free) • Chlorine Residual (Total) Supplier must use an ISO 17025 certified laboratory. Other parameters may be required depending on geographic area – contact OBO/OPS/SHEM.	Quarterly testing of 10% of bottles delivered for: • Total Coliform/e.Coli Post can test for coliform/e. Coli on-site using field test kits.	Based on supplier water quality results and Post QC Testing results

Trucked Water Sources

If trucked water is used for human consumption, SHEM will conduct an assessment based on water quality testing results. Post should verify the quality of Trucked Water sources prior to contracting with the supplier and then annually during the life of the contract. Once a contract is in place, additional quality control (QC) testing of each water delivery should be completed by Post (turbidity and free chlorine residual at a minimum; with total coliform/e.Coli, TDS, pH, and Temperature also recommended). These tests can be completed using an on-site water quality test kit with results entered into FAC Apps. Indicate the sample is from a trucked water delivery source and include the name of the supplier.

Post Managed Potable Water Treatment Systems – Potability Determinations

OBO Responsibility

OBO/PDCS/DE (PDCS) and OBO/CFSM/CM (CM) have the lead roles in potability determinations for new NEC/NCC potable water treatment systems. PDCS manages the planning and design tasks associated with the new potable water treatment systems. This includes evaluating source water quality data results and ensuring that the potable water treatment system is designed according to OBO codes, standards, and specifications, i.e. designed to produce water safe for human consumption.

CM has the lead role in the construction, transition, turnover, acceptance, occupancy, and operations phases of a new NEC/NCC, which includes the potable water treatment system. CM ensures that the potable water treatment system is constructed as designed by PDCS, and is commissioned and accepted according to OBO codes, standards, and specifications. CM has Policy Directives (PDs), Alerts, and Standard Operating Procedures (SOPs) in place that outline the processes used to ensure that the facilities (including the potable water treatment system) meet established performance and functional requirements prior to the Certificate of Occupancy being issued. These Policy Directives, SOPs, Alerts, and Specifications are listed below:

Reference	Title	Issued
Construction Management PD #01	Commissioning and Transition to Occupancy of Overseas Facilities	11/23/2021
Construction Management SOP 1	Commissioning	11/12/2021
Construction Management SOP 2	Transition to Occupancy	11/12/2021
Construction Management SOP 3	Certification Accreditation and Acceptance	11/12/2021
Construction Management SOP 4	Certificate of Occupancy	11/12/2021
Construction PD 02	Post Occupancy Warranty and Building Operations Inspection	4/18/2017
Construction Management Policy and Procedures Directive (PPD) CM 02	Post-Occupancy Warranty and Building Operations Inspection	4/18/2012

Reference	Title	Issued
Construction Alert A-2018-02	Construction to Occupancy Transition Process Improvement	March 1, 2018
Construction Alert A-2015-03	Integrated System Test (IST) for New Facilities	July 14, 2015
Construction Alert A-2016-01	Systems Manual Template	Feb 1, 2016
Construction Alert A-2016-02	Integrated System Test (IST) for New Facilities – Supplemental Information	March 16, 2016
Specification Section 017825	Operations and Maintenance Data	
Specification Section 017705	Closeout Procedures	
Construction Management Guide ²⁰	OBO Construction Management Guidebook	July 2023

CM's Commissioning Agent (CxA) prepares a project-specific Commissioning Plan that defines the specific commissioning tasks required to document the testing and performance of the various building systems (including the potable water treatment system). This Plan identifies the testing requirements and process for determining potability of the water produced by the potable water treatment systems prior to occupancy.

OBO/OPS/SHEM has the lead role in assessment and review of quality control (QC) monitoring results from ongoing operations of potable water treatment systems (POE and POU), analysis of treatment effectiveness from post treatment pilots, and following repair or service of water treatment, storage, and distribution systems. These assessments may include a review of operations and maintenance activities and/or water quality testing. This includes internally bottled water that is treated at their facility. SHEM (in coordination with OBO/FAC), assists Post in developing a Post specific OpMon/QC Plan and reviews the Post's QC testing data to determine potability.

OBO/OPS/FAC/PS (Water Testing Group) assists SHEM in their potability assessments by coordinating on development of OpMon/QC testing plans, providing training and technical support on sample collection and analysis, and training on use of the FAC Apps database. Potability assessments are also completed in coordination with the FAC Certificate-to-Operate program. Posts participating in this program will receive annual potability determinations in addition to tri-annual on-site sanitary surveys of their potable water treatment systems.

Post Responsibility

Posts develop and implement OpMon/QC Test plans for all potable water treatment systems, operate and maintain these systems, and enter all QC test results into FAC Apps.

²⁰ usdos.sharepoint.com/sites/OBO/OBO-Guide/CM-Guide/SitePages/CM-Guide.aspx

APPENDIX E: EXAMPLE OF COST COMPARISONS

Bottled Water vs Supplemental Treatment

In most cases, the comparison will be with a point-of-use (POU) water treatment device in a residence. We recommend identifying the expected useful lifetime of the water treatment. The following are some factors

Bottled Water	POU
Cost of water x annual consumption	Purchase cost
Cost of dispenser	Installation cost
Cost to operate dispenser	Electricity cost
Delivery	Replacement parts (e.g., filters)
QC Program (not required for NSF brands, etc)	Maintenance
	QC Program (not required for NSF brands, etc)

Distiller Water vs. Bottled Water

For the purpose of this comparison, cost data for equipment, bottled water, and electricity were compiled from bottled water vendors in the US and data from some Department overseas posts. Comparison period is for 5 years. Application is for Department overseas posts.

Assumptions.

- Family of four
- Consumption/use, per day, per person is 2 liters (drinking and cooking)
- Daily consumption/use for a family of 4 is approx 2 gals/day (4 X 2 liters = approx 2 gals)
- Annual consumption for a family of 4 is 2 gal/day X 365 days = 730 gals/year
- A five gal carboy of certified water delivered to a residence is approx \$5.00 or \$1.00/gal
- The cost of a cooler/dispenser is approx \$300.00 (includes the cost of electricity for one year).
- The operating lifetime of a cooler/dispenser is considered to be at least 5 years.
- Off-the-shelf bottled water costs approx \$1.50/gal.
- A high quality U.S. brand automatic distiller (e.g., Durastill, WaterWise, and B.J. David) produces 8 gals of distilled water every 24 hours (1 gal/3hours) and has a storage capacity of 4 to 5 gals. Unit costs approx \$800.00 plus \$50.00 shipping. Operating life/warranty is at least five years.
- Cost of electricity is approx 10 cents per KW hr
- A distiller uses approx 1000 watts/hr to produce 1 gal of distilled water in 3 hours.

Therefore, it takes 3 KWhr to make 1 gal water at a cost of \$0.30 (3 KWhr X \$0.10/KW hr).

Analysis of Annual Cost

PRODUCT	1 st YEAR	YEARS 2-5
Distiller 730 gals @ \$0.30/gal = \$219.00 plus \$850.00 (distiller purchase cost)	\$1,069	\$219.00
5 Gal. Carboy of Bottled Water & Dispenser 730 gals @ \$1.00/gal (carboy) = \$730.00 plus \$300.00 (dispenser/cooler purchase cost)	\$1,030	\$730.00
Off-the-shelf Certified Bottled Water 730 gals (annual use) @ \$1.50/gal = \$1,095.00	\$1,095	\$1,095.00

Assessment. After the first year (*BREAKPOINT*), the annual cost of distilled water is \$219.00 per year. Carboy water costs \$730.00 per year, and off-the-shelf bottled water costs \$1,095.00. Compared to 5 gal carboy water, there is an annual (2nd through 5th years) savings of \$511.00 per residence/per year (\$730 minus \$219). Compared to off-the-shelf bottled water, there is an annual (2nd through 5th years) savings of \$876.00 per residence/per year (\$1,095 minus \$219).

For the 5-year comparison period, the savings *per residence* for the 2nd through the 5th years after the *BREAKPOINT* (1st year) is a total of \$2,044.00 (\$511 X 4 years) for the distiller vs. 5 gal carboys with cooler/dispenser, and \$3,504.00 (\$876 X 4 years) for the distiller vs. off-the-shelf bottled water.

Cost Savings for the Four-Year Period After Breakpoint

	Distiller vs. 5 Gal Carboys of Bottled Water	Distiller vs. Off-the-shelf Bottled Water
25 RESIDENCES	\$51,100.00 (25X\$2,044)	\$87,600.00 (25X\$3,504)
50 RESIDENCES	\$102,200.00 (50X\$2,044)	\$175,200.00 (50X\$3,504)
75 RESIDENCES	\$153,300.00 (75X\$2,044)	\$262,800.00 (75X\$3,504)
100 RESIDENCES	\$204,400.00 (100X\$2,044)	\$350,400.00 (100X\$3,504)

Addendum. The cost of a 5-gal carboy of bottled water and the cost of off-the-shelf bottled water products are intended to be representative. Actual bottled water costs may vary from post to post. Cost comparisons can be adjusted by using actual bottled water cost data and following the same paradigm.

APPENDIX F: WATER TREATMENT SYSTEMS

Selecting effective water treatment for post is complex. Water treatment should be chosen based on ability to remove and/or disinfect identified contaminants in the water. There are additional considerations, such as water production rate, cost, and ease of maintenance, but these considerations are secondary to ensuring that water treatment can remove the identified contaminants.

Approaches to Water Treatment

There are various methods for purifying water. Appropriateness depends on the type and degree of contaminant removal needed. The types of methods used to purify water frequently include the following either alone or in combination.

Boiling

In order to disinfect water by boiling, it is important to filter it first. Filtering removes particulates (sometimes invisible) which provide a physical barrier to protect microorganisms from destruction. A three-minute rolling (or full) boil will effectively kill all microorganisms including Hepatitis A virus with a margin of safety to allow for the fall in boiling point as a function of altitude (one degree Fahrenheit for each 550 feet of elevation). Protozoan cysts and bacteria are devitalized after a one-minute rolling boil. Following boiling, water should be either allowed to cool in the same vessel or in a clean container.

Do not filter the water after boiling, as this can contaminate the water. For cryptosporidium (a protozoan), boiling will be a better choice for disinfection than halogens as this organism and its cysts are halogen resistant.

Boiling is not an effective treatment method for inorganic compounds such as metal (iron and lead) and chemical contamination.

Filtration

Filtration physically removes particulates, which include sediments, some heavy metals, and microbiological agents. The ability of a filter to remove various contaminants depends on the filter pore size. The smaller the pore size the more complete the filtration, but the longer the process and the more quickly the filter may clog. Removal of these components improves the efficacy of subsequent disinfection by halogens such as chlorine or extreme heating.

A small pore size of less than 1 micron can effectively remove most microorganisms including giardia and cryptosporidium, which resist disinfection by chlorine and other halogens. Ceramic filters (e.g. candle type, porcelain Pasteur or Berkefield, have an effective pore size of 0.20 microns which filter out most microorganisms except certain enteric viruses such as Hepatitis A (0.027 micron size), which are too small. Viruses, however, are easily killed by chlorination or

heat.

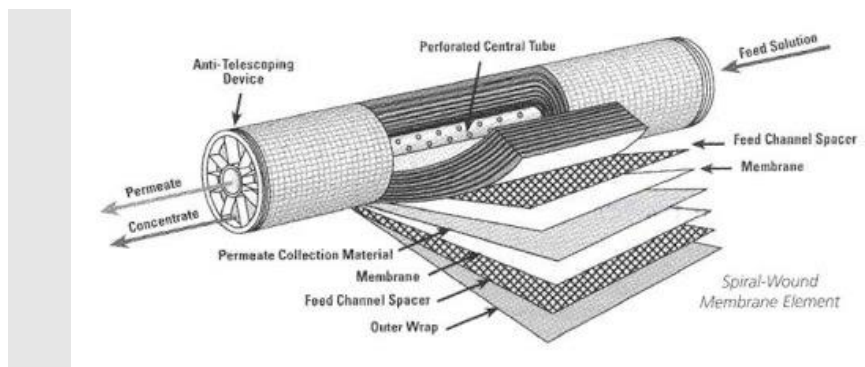
Filters are typically classified by their pore sizes. The different types of filters and their pore sizes are outlined below:

- Microfilter: $10\mu\text{m}$ - $0.1\mu\text{m}$
- Ultrafilter: $0.1\mu\text{m}$ - $0.01\mu\text{m}$
- Nanofiltration: $0.01\mu\text{m}$ - $0.001\mu\text{m}$

Reverse Osmosis (RO)

Reverse osmosis systems purify water by rejecting substances in the water based on their molecular properties and the type of RO membrane. The contaminated water passes through a semi-permeable membrane that passes water molecules but blocks most dissolved and suspended molecules. The natural direction of flow (osmosis) from the product side of the membrane to the feed side is reversed by applying enough pressure to the feed water to overcome its loss of osmotic potential. Hence the term “reverse osmosis”.

Reverse osmosis can remove ions, turbidity, bacteria, viruses, and cysts. However, Reverse Osmosis membranes may reject 70% or more water which drains off as wastewater. Raw water parameters such as pH, temperature, and turbidity can affect membrane performance. To preserve the membranes, pre-filtration is typically added to a reverse osmosis system.



Reverse Osmosis Membrane

Distillation

Distillation is a process of heating the water into steam then re-condensing the steam to produce purified water. Dissolved minerals (inorganic compounds), nonvolatile organics, and particulates do not evaporate with the water and are left in the boiling chamber. The steam is then cooled and condenses to form purified water. Distillation is most effective in removing inorganic compounds such as metals (iron and lead) and nitrates, hardness which is mainly calcium and magnesium and, particulates from contaminated water. The extreme heat provides thorough disinfection, killing bacteria, viruses, and cysts.



Image of a Durastill distiller

Activated Carbon

Activated carbon removes contaminants by adsorption. Activated carbon is particularly effective in removing organic compounds, such as VOCs and THMs, which adhere to the carbon. Activated carbon suited for water treatment is made from bituminous coal, peat, lignite, coke, wood, and coconut shells. In order for the carbon to remove the contaminants from the water, there must be sufficient contact time with the water. Eventually, the capacity of the carbon is exceeded and no further removal will occur. The carbon must be replaced. Activated carbon is typically incorporated inside multi-stage POU treatment systems.

Ion Exchange

Ion exchange works by exchanging charged ions in the water with ions of similar charge from the ion exchange resin. The type of resin determines whether there will be a cation exchange or anion exchange. In cation exchange, most often sodium ions are added to the drinking water as other positively charged ions such as calcium and magnesium are removed. Cation exchange is primarily used for water softening and the sodium levels that result may create dietary concerns for individuals on reduced sodium diets. In anion exchange, negatively charged ions such as nitrate are removed and chloride is added to the water. After a period of time, the exchange resin becomes saturated with contaminant ions and must be regenerated.

Ion exchange can also be used to remove metals such as lead, arsenic, and hexavalent chromium, along with sulfate.

Chlorination

Chlorine is an effective and widely used disinfectant and oxidant, but is more suitable for POE

systems, large treatment plants, or manual introduction, and is not used in POU devices. Pre-filtration (≤ 1 micron) is necessary to physically remove cysts that are resistant to halogens such as chlorine.

A chlorine residual concentration (0.2 to 0.5 ppm) provides continuing protection from microbiological agents and can be checked at regular intervals (e.g. daily).

Ozonation

Ozone, an unstable form of oxygen and a strong oxidizing agent, disinfects well and oxidizes inorganics and organics. It is used primarily in community water treatment plants and sometimes in point-of-entry (POE) systems. Ozone's effectiveness against cysts is still being researched. It is unlikely that home or personal POU treatment units will employ ozone. The OBO Plumbing Code currently prohibits the use of Ozone disinfection as a potable water treatment method.

Ultraviolet Light

Certain wavelengths of ultraviolet (UV) light effectively sterilize microbes, preventing them from replicating in a host. Turbidity in the water can shield microbes from the effects of UV light, so UV devices are only effective when the raw water is always clean, clear, not subject to chemical or organic pollution, and operated as intended. Turbidity, color, dissolved iron and organic compounds diminish performance. Prefiltration may be required to remove these characteristics. UV devices also require a constant and reliable supply of power. Because of these factors, UV treatment is rarely the best choice for water treatment. OBO current Plumbing Code (2022) prohibits the use of UV disinfection as a potable water treatment method.

NSF/ANSI 55 certification is the standard for UV water treatment. There are two classes: Class A systems which are designed to disinfect and/or remove microorganisms, including bacteria, viruses, *Cryptosporidium* oocysts and *Giardia* cysts from contaminated water to a safe level. Class B systems are used to provide supplemental treatment to water that is already fit for human consumption.

We recommend procuring only NSF/ANSI 55 certified Class A treatment devices.



Ultraviolet light water treatment device

Point of Entry (POE) and Point of Use (POU) Water Treatment

Supplemental Water Treatment technologies can be classified as either Point of Entry (POE) or Point of Use (POU). Each type is explained in detail in the sections below:

Point of Entry (POE) Systems

POE systems treat the raw water that comes into a facility, so it provides treated water throughout the entire building or compound. POE systems are typically found at the Embassy compounds and provide safe drinking water at taps and water fountains. POEs are not exclusive to the Chancery however. Some posts have “whole house” POE systems installed at residential compounds and individual residences.

During New Embassy Compound (NEC) construction, POE systems are designed and installed using a formal process based on OBO’s Codes, Standards, and Specifications. These systems typically use a series of treatment methods, such as filtration and chlorination, and are designed by architectural and engineering (A&E) firms based on the results of water testing of the raw water.

POE systems are formally commissioned with the facility staff at Post.

Posts may also participate in FAC’s Certificate to Operate (CTO) auditing and maintenance program during its lifecycle operation. This program goes beyond the commissioning process and assists Posts with ongoing operations and maintenance by helping ensure that: 1) operators are trained, 2) adequate chemicals and media are available, and 3) operational and quality control monitoring (OpMon/QC Test Plan) is in place to verify the system is functioning as designed and the treated water remains fit for human consumption.

Questions about POE systems should be directed to ***OBO/PDCS/DE (PDCS) and OBO/OM/FAC/PS Water Testing Group.***



Image of a Reverse Osmosis water treatment plant

Point of Use (POU) Treatment Devices

In contrast to POE systems, POU devices treat only the water at the point of consumption and are typically located at the tap or under the sink. These devices can be very practical and efficient since they purify only the water taken from the tap rather than all the water in the building.

When selecting a POU water treatment device or determining whether or not a POU currently in use is appropriate, the most critical consideration is whether it will effectively remove the contaminants of concern at concentration levels present in the source water. In order to make this determination, the contaminants of concern must be identified and the treatment technologies employed by the POU and their removal efficiency evaluated.

All POU devices require periodic maintenance and monitoring to ensure that they continue to function effectively. Periodic maintenance may involve:

- replacing filters and membranes on a scheduled basis for cartridge type purifiers and reverse osmosis units
- cleaning and descaling boiling chambers in distillers
- scrubbing, then sterilizing ceramic microfilters on a regular basis
- replacing and/or keeping lamps clean on UV devices so they remain effective

OBO (OPS/SHEM and OPS/FAC/PS) can assist in the review of the appropriateness of POU's. Be certain to gather and provide all pertinent data on the local water quality and the POU's being considered.

References

Primary source for above water treatment technologies are 1) Point-of-use/Point-of-entry for Drinking Water Treatment, book by B. Lykins, R. Clark, and J. Goodrich (EPA Water Scientists), Lewis Publishers, Inc, 1992. And 2) Point-of-Use or Point-of Entry Treatment Options for Small Drinking Water Systems, EPA 815-R-06-101, April 2006²¹

Selection and Installation of Water Treatment

In the absence of potable water from a municipal or Post source, the source water may be treated to meet standards using:

- Point of use (POU) treatment devices, and/or
- Point of entry (POE) treatment systems

²¹ https://www.epa.gov/sites/default/files/2015-09/documents/guide_smallsystems_pou-poe_june6-2006.pdf

Point-of-Use (POU) Water Treatment Devices

As described by the name, POU devices are typically located at the tap or under the sink. These devices can be very practical and efficient since they purify only the water taken from the tap rather than all the water in the building. The water is also typically consumed shortly after treatment, which means addition of residual protection (drops of chlorine) is not necessary most of the time. Successful POU Treatment operation demands the following:

- Appropriate selection of device
- Proper O&M including Operational Water Quality Monitoring (See Appendix G)
- Quality control verification (see Appendix G)

Selection of POU Devices. When selecting a POU water treatment device or determining whether or not a POU currently in use is appropriate, the most critical consideration is whether it will effectively remove the contaminants of concern at concentration levels present in the feed water. In order to make this determination, the contaminants of concern must be identified and the treatment technologies employed by the POU and their removal efficiency evaluated.

Additional important considerations include:

- Energy requirements
- Water pressure requirements
- Space requirements
- Care and maintenance
- Ease of maintenance and availability of components
- Reliability and durability.
- Heat emission

The Department requires that POU devices be independently certified by a third party certification agency to remove the identified contaminants of concern and/or that the POU devices be piloted in the field at a small number of locations before wide-scale installation.

Third Party Certification. Third party agencies that certify POU Treatment Devices include NSF, WQA (Water Quality Association), or UL (Underwriter's Laboratory). These agencies test POU devices according to established standards developed by NSF/ANSI. Standards for POU Treatment Devices include:

NSF/ANSI 42 – Aesthetic and material integrity
NSF/ANSI 53 – Contaminants of Health Concern

NSF/ANSI 58 – Reverse Osmosis Systems
NSF/ANSI 62 - Distillation

Posts should request verification of third party certification from the supplier prior to purchase of any POU device, and verify that certification through the third party certifying agency. OBO (OBO/OPS/SHEM and OBO/CSFM/FAC FAC Water Testing Group) can assist with the verification.

Do not confuse any EPA registration and establishment number on the label of a POU as a certification of quality. This registration refers to the disinfectant employed, which is technically a pesticide and must be registered if sold in the US. Disinfection devices using ozone or ultraviolet light are not required to be EPA registered.

Pilot Procedures for POU Devices. POU treatment devices designed to provide water safe for human consumption should be piloted at a subset of locations before installing them throughout a facility or residential housing pool. Post should contact OBO (OBO/OPS/SHEM and OBO/CSFM/FAC/PS Water Testing Group) for assistance with the pilot study, which will include:

- Evaluation of source water quality to determine the appropriate certified POU treatment device (Post and OBO).
- Installation and testing of the POU treatment device at selected locations (Post)
- Evaluation of testing results to determine full-scale implementation (OBO)

Full-scale Implementation of POU Devices. If the pilot study determines that a POU treatment device is acceptable, they may be installed full-scale. O&M procedures must be established, including ensuring GMMS work orders are in place for routine maintenance and developing and implementing an OpMon/QC Testing Plan,

Point of Entry (POE) Water Treatment

Posts sometimes treat water as it enters a property using a point-of-entry (POE) water treatment system. POE systems are attractive because they can treat all the building water rather than just the water at the tap. POE water treatment system can produce water safe for human consumption and also include treatment to control scaling and corrosion that prolongs the life of plumbing and prevents harm to building systems that use water. Operation of a POE system is basically a small water treatment plant that requires proper operation and maintenance in order to assure the quality of the finished water. Successful POE system operation demands the following:

- Appropriate design
- Proper installation and commissioning

- Trained Operators (see Appendix G)
- Proper O&M including Operational Water Quality Monitoring (see Appendix G)
- Quality Control (QC) verification (see Appendix G)

POE systems combine various components (e.g. filters, softener, chlorinator) to achieve the desired water quality. Some are components packaged into POE units purchased commercially and installed with little or no modifications. As with POU's, the starting point for designing and/or selecting a POE system is determining the quality of the local water supply, both in terms of contaminants with health consequences and aesthetic properties. This information is key in determining the degree of treatment that will be provided and whether the POE will be designed to provide potable water. If the POE does not provide potable water, then POU's and/or bottled water will be needed.

Design and Installation of POE Water Treatment Systems. POE Water Treatment Systems should be designed to meet the Department's OBO Plumbing Code²² and standard specifications for potable water treatment design and construction.²³ The code and specifications identify the quality of water that should be produced by any POE water treatment system, including maintenance of a chlorine disinfectant residual of between 0.2 – 0.5 mg/L everywhere in the distribution system; TTHM/DBP removal to less than 0.1 mg/L; turbidity of less than 0.5 NTU; a positive Langelier Saturation Index (LSI); and where local water or well water has measured hardness level (Calcium Carbonate) of greater than 170 mg/L, providing water softening system to reduce hardness level to between 50 and 85 mg/L.

POE Pilot Procedures. (to be completed in future versions)

Commissioning for New POE Systems or for Repairing existing POE systems. (to be completed in future versions)

²² OBO-ICD 2020 OBO – International Codes Supplements, Amendments to the International Plumbing Code, January 2020 (Chapter 3, 312.11 and 312.12; Chapter 6, 601.4 (tests),

²³ United States Department of State Overseas Buildings Operations OBO Standard Specifications, Section 223230 – Potable Water Treatment. January 2018R.

APPENDIX G: GUIDANCE ON ROUTINE O&M FOR POTABLE WATER TREATMENT SYSTEMS

POU Treatment Devices

O&M of POU Devices

All POU devices require periodic maintenance and monitoring to ensure that they continue to function effectively. Periodic maintenance may involve:

- replacing filters and membranes on a scheduled basis
- cleaning and descaling boiling chambers in distillers;
- scrubbing, then sterilizing ceramic microfilters on a regular basis; and
- replacing and/or keeping lamps clean on UV devices so that it remains effective.
- Conducting operational monitoring, entering results in FAC Apps, and adjusting O&M based on results as needed.

Post Facilities staff should ensure that O&M checklists for all manufacturer's required maintenance activities for POU treatment devices have been entered into GMMS, and work orders established at the recommended frequencies. Operational water monitoring is a required maintenance activity

Quality Control (QC) Testing Verification of POU Devices

POU devices used for human consumption must have a QC testing plan in place, with results entered into FACApps. SHEM will review results and verify potability (see Appendix D).

POE Water Treatment Systems

O&M of POE Water Treatment Systems and Internally Bottled Water

Effective operations and maintenance of POE Treatment Systems is needed to ensure quality water is continuously produced. POE systems have far greater capacity and more critical operating elements than POU devices, thereby requiring more specialized training on the part of the responsible operating personnel at post. The Facilities staff at post should maintain oversight of the operation and maintenance of the POE system to ensure that the system operator(s) is following all the required operating instructions. This includes ensuring that all O&M checklists for required maintenance activities for POE treatment equipment assets, water storage tank cleaning and disinfection²⁴, distribution system maintenance (backflow prevention, valve exercising valves), and operational and QC water quality testing have been entered into GMMS, and work orders established at the recommended frequencies.

Water that is treated by post and then bottled and distributed throughout the compound or to

²⁴ Contact OBO/CFSM/FAC Water Testing Group for guidance on water storage tank cleaning and distribution system maintenance activities

residences, must follow stringent hygiene controls and the quality control requirements. Standard operating procedures must be written and individuals bottling the water must be trained on proper hygienic procedures. Guidance is provided below for the use of drinking water dispensers and bottles*.

QC Verification of POE Water Treatment Systems

All Posts with POE Water Treatment Systems (including any internally bottling of POE treated water) should have a QC monitoring plan in place to verify safety for human consumption (potability). SDEM assesses the potability of POE treated water through evaluation of results from ongoing QC monitoring completed by Post and entered into FAC Apps, and comparison of these results with Department standards (Appendix A). Potability assessments are also completed in coordination with the FAC Certificate-to-Operate program. Posts participating in this program will receive annual potability determinations in addition to tri-annual on-site sanitary surveys of their potable water treatment systems.

Trucked Water

In addition to the supplier's verification of the water quality and Post QC testing of the deliveries, the following guidelines should be used to help ensure the safety of trucked water:

1. Supplier should identify the source of the trucked water
2. Truck container should be contaminant-free, watertight, and made of a water-grade approved material
3. Truck container should only be used to haul water for drinking or other liquid food grade products. Any tanks used to transport chemicals, petroleum, etc... shall not be used to haul drinking water. All equipment associated with the trucked water (collection, transport, and delivery) should only be used to transport drinking water or liquid food grade products.
4. Tanks should be completely sealed with secured openings.
5. Tanks, as well as hoses, pumps, and other equipment used to transfer the water should be cleaned and disinfected prior to use
6. Tanks should be filled and emptied using an air gap to prevent backflow and contamination of the water in the tank
7. Post should maintain records of each trucked water delivery, including:
 - a. Name of the supplier
 - b. Quantity delivered
 - c. Water source used
 - d. Date and time of delivery
 - e. Free chlorine residual and turbidity measured at the point of delivery
 - f. Confirmation from supplier that proper disinfection was performed

Drinking Water Dispensers/Coolers, and Bottled Water Safety

At some overseas posts, the locally available municipally treated water is safe for drinking at

the tap. When local water is not safe for drinking from the tap, posts either centrally treat local water to achieve potability by using point-of-entry (POE) systems, or point -of-use (POU) devices at sink locations, or resort to purchasing or producing bottled water. These guidelines focus on precautionary measures which should be implemented to ensure potability of bottled water and protect against contamination in drinking water coolers/dispensers, and from contamination which can occur in the transference of water from a potable source to sanitized containers such as five gallon carboys (jugs). Procedures for doing this are described below. The ergonomic hazards of transporting heavy containers of water cannot be ignored. Alternatives to bottling drinking water include installing point-of-use (POU) devices at various sink locations. Posts should develop tailored written operating procedures and provide training on POU devices, when used.

Personnel Sanitation

As "food" handlers, individuals bottling drinking water must first wash their hands with soap and water and dry their hands with disposable paper towels. The use of rubber gloves is recommended. The post health unit should be contacted for further guidance on personal hygiene of food handlers. Precautions are needed to guard against contamination of the water from unsanitary bottles, bottle exteriors, and from loading bottles into cooler/dispensers.

Bottled Water Scenario

SHEM is aware of the following bottled water practices which occur at a number of posts singularly or in combination.

1. Five-gallon jug (carboy) filling station. Water emanating from a potable water source such as a purification system (e.g., Reverse Osmosis (RO) system) is directed to a filling station via a dedicated pipeline. Five-gallon plastic jugs are filled with potable water using a sanitary filling device such as a preferred spring supported filling wand similar to those used in food service installations. (See the NSF International (NSF) website. www.nsf.org. NSF contact is Jeremy Brown. Tel (734) 769-8010, FAX (734) 769-0109.) Once the five-gallon jugs are filled, they are transported to locations where they are inverted into drinking water dispensers/coolers replacing the empty jug.
2. Locally procured five-gallon jugs of potable water. Many post purchase potable drinking water in 5 gallon jugs from local companies for use in drinking water dispenser/coolers both in offices and residences. Posts should ensure that the purchased water is potable based on laboratory test documentation provided by the supplier and/or by periodic testing for coliform bacteria by the post health unit. In this scenario, the empty five-gallon jugs are replaced by filled jugs by the supplier providing this service on a scheduled basis

NOTE 1: It is recommended to use only bottom fill dispensers (not top fill). Periodic cleaning of a water dispenser/cooler is required. At a minimum, cleaning should be performed on a monthly basis. See cleaning and disinfection procedures under the "Water Storage" section in these guidelines. Consult the dispenser operations manual for specific instructions. Generally, these instructions will entail filling the dispenser with a disinfecting solution (e.g., 2 tablespoons (30ml) of unscented household bleach (5.25% available chlorine) such as Clorox, per gallon of potable water, which equates to 400ppm), allowing the disinfecting solution contact time (approximately 15 minutes), draining the disinfecting solution, and, finally flushing with potable water before refilling. Note that some water dispensers have hot water reservoirs/tanks and instructions may contain caution against using bleach for cleaning.

NOTE 2: Personnel handling filled five-gallon water jugs prior to placing them in a dispenser should wash the outside of the bottle with soap and water and dry the bottle with paper towels. Using paper towels dipped in a solution of 1 teaspoon of bleach in 1 liter of water, wipe the outside of the bottle paying particular attention to the cap and neck of the bottle. Start at the cap and work down. Dry the bottle with paper towels.

NOTE 3: Five-gallon jugs of water are quite heavy. When placing the jugs into the dispenser, personnel should wear food handler's gloves. Carts for transporting the containers and training on lifting techniques should be provided.

Filling containers at a central potable water source. At some posts, potable water (e.g., from a water purifier such as a central POE, or a large capacity distiller with a large capacity reservoir centrally located in the chancery) is made available to personnel who can draft the potable water into containers (e.g., 1 gal plastic bottle) and transported to their home (or office) for drinking and cooking uses. To ensure the safety of the water for drinking, it is essential that the instructions for maintaining product (such as a distiller) sanitation provided by the manufacturer are carefully followed. Personnel using their own containers, such as plastic bottles, must practice good personal hygiene and sanitation activities (e.g., washing containers with soap and water and thoroughly rinsing with potable water) to avoid bacterial contamination of the bottle water. To avoid contaminating drinking water while filling the containers or drawing water for consumption, pour or drain the water from containers rather than ladling. To improve the taste of the water (reduce the chlorine taste and add oxygen to stale water) pour the water back and forth between two

clean containers or add a pinch of salt.

Water Storage

Store drinking water in containers with lids or some other covering. Sound plastic, unbreakable containers are preferred. Containers and their lids must be THOROUGHLY cleaned and disinfected prior to filling. If the containers are visibly dirty, the interior should be scrubbed with a brush. Clean the bottles with a small amount of soap. A disinfecting solution, such as one prepared by 3 tablespoons (about 44ml) of household bleach (assuming the % of sodium hypochlorite is 5.25%, this would make a 600 ppm concentration) per gallon of water, should have contact with the entire surface, then drained. Do not use household bleach that contains detergent or fragrance. Finally, the container should be rinsed with potable water and, if necessary, air dried.

If the water will be stored for more than a day and is not already chlorinated (must have a free chlorine residual of at least 0.2 ppm), add 2 to 4 drops of household bleach (assuming the % of sodium hypochlorite is 5.25%) per quart (or liter) of water or 1 teaspoon per 5 gallons. Seal the container tightly, label as “drinking water” and store in a cool, dark place until ready to use.

Note: Posts using bottled water should ensure that the product meets one of the following:

- Certified by the NSF. Products include international brands. See website www.NSF.ORG.
- From a reliable source e.g., US military certified bottled water, international brand, brand used by an airline, or produced locally and certified by credible laboratory test data.
- Produced internally and satisfies the safety measures contained in these guidelines.

Technical Assistance. OBO/OM/SHEM should be contacted for technical assistance on the safety of drinking water dispenser/coolers, bottled water, and POU water treatment devices.

APPENDIX H: GUIDANCE ON INCIDENCE RESPONSE FOR WATER SYSTEMS

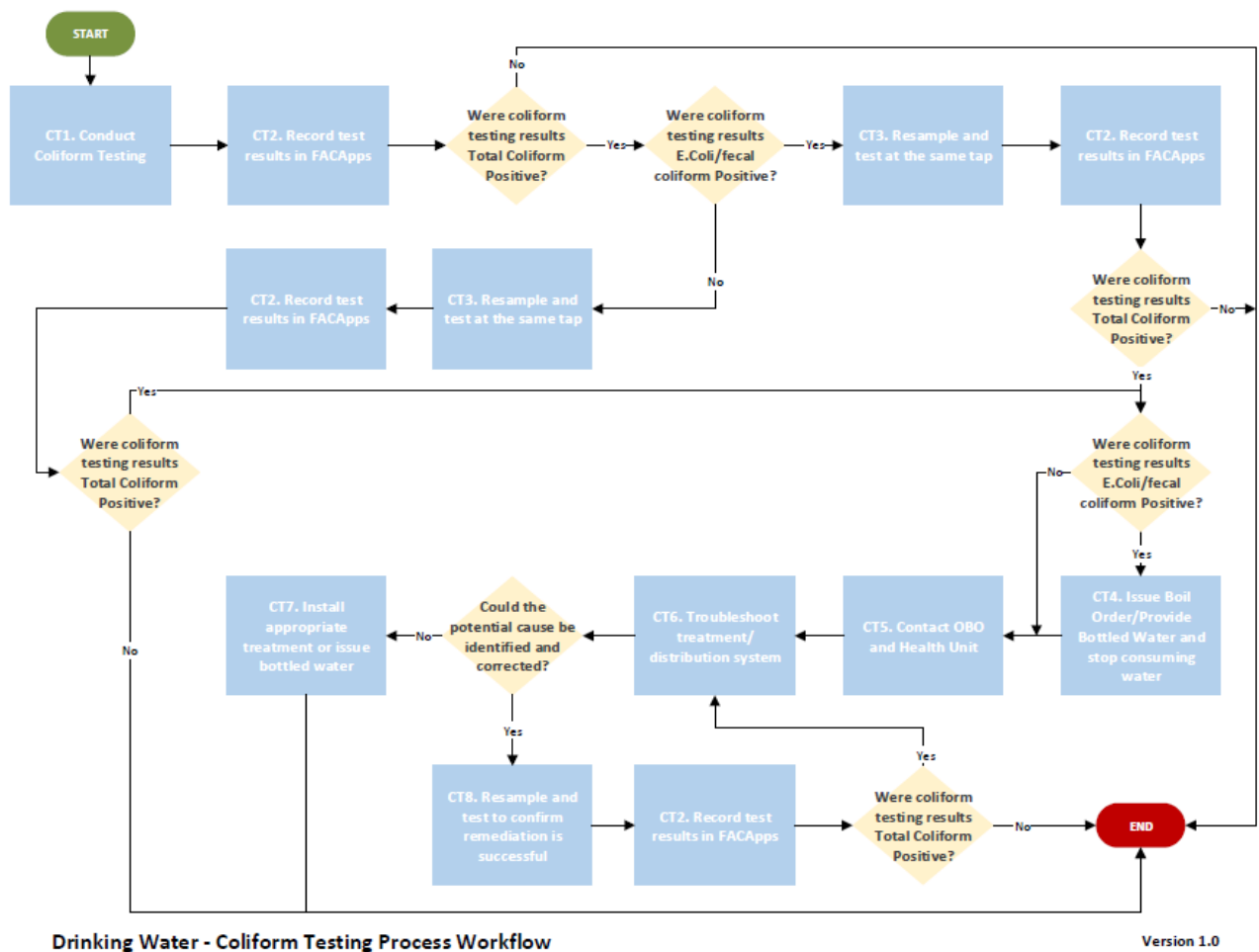
Exhibit H-1: Microbiological Contaminants

Coliform Testing Process Workflow

This document expands on the process steps define in the Drinking Water - Coliform Testing Process workflow. This is intended to be general guidance and the process steps documented here may change depending upon specific circumstances at post.

Please note that this process applies to potable water only. Although the testing steps may be similar for water used for contact purposes (showering, bathing, etc.), responses such as issuing a “Boil Order” apply only to water used for consumption.

Please contact Justin Omps at SHEM (ompsj@state.gov) and the FAC Water Testing Group (Water_Testing_Group@state.gov) if you have any questions or concerns.



I. CT1. Conduct Coliform Sample Collection and Testing

Sample water from the location specified in the water testing plan. Because it is easy to contaminate a sample and create a false positive result, follow the sampling protocol steps outlined below to take the sample:

- I. Wash your hands thoroughly with soap and water or use a disinfecting wipe prior to

beginning the sampling procedure.

- II. Remove any attachments from the faucet, including aerators, screens, washers, hoses, and water filters. Disinfect the sample site before collecting the sample with an alcohol swab or bleach, be sure to flush the site thoroughly to remove all disinfectant.
- III. Turn on the cold water only and let it run with a steady stream for at least 3 (three) minutes. Before collecting the sample, turn the water down to a thin stream (about the width of a pencil), then let the water run one minute. If you chlorinate your system, measure the free chlorine residual and note the measurement.
- IV. To avoid contamination while taking the sample, hold the bottle near the bottom with one hand, hold the top of the cap with the other, and then unscrew the cap.

Note: If using a sampling kit, there may be some liquid or powder in the sample bottle to neutralize chlorine. Do not rinse it out.

Note: Do not set the cap down, touch any part of the cap that touches the bottle, or let anything touch the rim of the bottle or the inside of the cap.

- V. Hold the bottle under the stream of water. Be careful not to let the bottle touch the sample tap. Fully fill the bottle to the neck or indicated fill line. Don't allow it to overflow. Remove the bottle from the water flow and carefully screw the cap back on.

All coliform samples should be tested for both Total Coliform, and if positive, for E.Coli/Fecal Coliform. The test sample must be incubated. Post¹ can run the test on-site using a coliform test kit (Colisure, Colilet, or Colitag) and an incubator, or through an outside certified laboratory.

¹ Facilities and/or Health Unit staff may collect, analyze, and report coliform testing results. Posts should identify responsibilities for sample collection, testing (analysis), and reporting as part of their Drinking Water Safety Program.

- a. Certified Lab Testing: transport samples in a cooler with an ice pack, to a certified lab or a designated drop-off location, as soon as possible. Lab testing (analysis) must begin within 30 (thirty) hours after you collect your sample.
- b. Post On-site Testing: For larger posts, transport samples in a cooler with an ice pack, to the Post testing location (Health Unit, Facilities, etc...) as soon as possible.

Generally, samples should be tested within 6 hours after sample collection and within 2 hours from receipt of sample at the Post testing location, but please follow specific manufacturer instructions for your coliform testing kit.

- VI. Deliver the samples for testing:

II. CT2. Record Test Results in FACApps

Post personnel should enter the results of all water quality tests and retests in FACApps, regardless of results. Use the link below to access the FACApps Water Quality Testing module where you can enter the test results.

<https://httpfacappsobostatesbu8161homeaspx-usdos.msappproxy.net/FACTabs.aspx?appid=9#s=Modules%2FWQT%2FHome.ascx>

Note: The Water Quality Testing module is a user authenticated module and requires the user to have been granted prior access. If you do not have access, click the Request Access button on the FACApps homepage.

1. Click the **Reading** tab at the top of the screen.
2. Click **Submit Readings**.
3. Fill in the appropriate fields and click the **Save and Continue** button to move to subsequent

pages.

Note: Please ensure that you add the RPA Prop. ID to the Building field and that the Location field contains sufficient information to identify where the sample was taken (source water, water tanks, pre-treatment, post-treatment, etc.)

III. CT3. Resample and Test at the Same Tap

Because sample contamination is common, all positive tests should be retested as soon as possible at the same tap, to confirm the presence of coliform bacteria. Follow the same sampling procedures listed in CT1. Conduct Coliform Testing. Alternatively, if sample contamination is a concern, request that Health Unit personnel perform the water sampling for the retest.

Depending upon the test results, there are 3 possible courses of actions:

- 1. Total Coliform – (negative) and E.Coli – (negative):** This indicates that the original positive result was a false positive. Post should continue normal monitoring procedures.
- 2. Total Coliform + (positive) and E.Coli – (negative):** Coliform contamination confirmed but there is no fecal coliform indicators present. This does not require a boil order and residents do not have to stop consuming the water, but Post should contact OBO (FAC and SHEM) and begin troubleshooting the source of the contamination.
- 3. Total Coliform + (positive) and E.Coli + (positive):** Fecal coliform contamination is confirmed. Post should issue a boil order to affected personnel (see template) and contact OBO (FAC and SHEM). Post should begin troubleshooting the source of the contamination and follow the process outlined in this document for confirmatory testing to put the water system back into service.

IV. CT4. Issue Boil Order/Provide Bottled Water and Stop Consuming Water

If a retest sample confirms the presence of E.Coli/Fecal Coliform, Post can no longer consider the water potable. Post should issue a Boil Order notification to affected residents or personnel and if practical, issue bottled water or use an alternate water source. Use the Boil Order template attached to this document. Boil Order notifications should be communicated to post management, OBO, and the Health Unit (See **CT5. Contact OBO and Health Unit**).

After the source of contamination has been identified and fixed, collect three (3) samples on three (3) separate days at the same location where the initial sample was confirmed positive for coliform, (See **CT8. Resample and Test to Confirm Remediation is Successful**). The boil order should remain in effect until all 3 samples are negative for Total Coliform, indicating that the water is free of contamination.

V. CT5. Contact OBO and Health Unit

Post personnel should report all confirmed positive coliform tests to the post Health Unit, the RMLS, and notify Overseas Building Operations (OBO). Post personnel should send an email to the OBO FAC Water Testing Group (Water_Testing_Group@state.gov) and to Justin Omps at OBO/SHEM (ompsj@state.gov). When reporting results to OBO, please include detailed information about the test results and sampling location. OBO personnel will contact post to provide further assistance.

VI. CT6. Troubleshoot Treatment/Distribution System

OBO will provide post with troubleshooting and water treatment system guidance to address the source of coliform contamination. Post follows the guidance to address the source of coliform

contamination. OBO will work with post personnel to ensure that proper cleaning and disinfection methods are used to clean water tanks, shock distribution systems, repair decencies, and return the water system to service.

VII. **CT7. Install Appropriate Treatment or Issue Bottled Water**

If post personnel are unable to address the source of contamination (problems with municipal supply or distribution, landlords at leased properties unresponsive, etc.) post should work with FAC and SHEM to identify and install appropriate water treatment or provide bottled water. It is important that post personnel select water treatment that is certified to remove the contaminants. Post should work with SHEM and the FAC Water Testing Group to select treatment rather than purchasing treatment devices that are available on the local market. If issuing commercial bottled water, SHEM recommends water that is International Bottled Water Association (IBWA), National Science Foundation (NSF), or DoD certified. SHEM can also assist with the necessary justification memos for posts Financial Management Officer (FMO).

VIII. **CT8. Resample and Test to Confirm Remediation is Successful**

After the source of contamination has been identified and fixed, collect three (3) samples on three (3) separate days at the same location where the initial sample was confirmed positive for coliform. Follow sampling procedure in **CT1. Conduct Coliform Sample Collection and Testing**.

Boil Water Template

BOIL WATER NOTICE

_____’s _____ Tested Positive for Fecal Indicator

BOIL YOUR WATER BEFORE USING

Our water system recently detected a fecal indicator (E. coli) in the water provided for human consumption (See *Note Below). You have a right to know what happened and what we are doing to correct this situation.

On _____, we collected a sample from _____.
The sample was confirmed to be positive for E. coli.

What should I do? What does this mean?

DO NOT DRINK THE WATER WITHOUT BOILING IT FIRST. Bring all water to a boil, let it boil for three minutes, and let it cool before using, or use bottled water. Boiled or bottled water should be used for drinking, making ice, brushing teeth, washing dishes, and food preparation until further notice. Boiling kills bacteria and other organisms in the water.

What is being done?

We will inform you when tests show no E. coli and you no longer need to boil your water. We anticipate resolving the problem within _____.
For more information, please contact _____ at _____.

Please share this information with all the other people who drink this water, especially those who may not have received this notice directly (for example, people in apartments, nursing homes, schools, and businesses). You can do this by posting this notice in a public place or distributing copies by hand or mail.

This notice is being sent to you by _____.

Date distributed: _____.

***NOTE:** Fecal indicators are microbes whose presence indicates that the water may be contaminated with human or animal wastes. Microbes in these wastes can cause short term health effects, such as diarrhea, cramps, nausea, headaches, or other symptoms. They may pose a special health risk for infants, young children, some of the elderly, and people with severely compromised immune systems. These symptoms are not caused only by organisms in drinking water. If you experience any of these symptoms and they persist, you may want to seek medical advice. People at increased risk should seek advice from their health care providers about drinking this water.

Exhibit H-2: Nitrate, Lead, Arsenic Contaminants

(create process flow similar to coliforms – for future version of document)

Exhibit H-3: Disinfection By-Products

(create process flow similar to coliforms – for future version of document)

Exhibit H-4: Other Regulated Parameters

(Inorganics, Organics, Rads)

(create process flow similar to coliforms – for future version of document)

Exhibit H-5 Other Contamination Events

(for future version of document)

Source contamination

Fuel Spill

Other

Distribution contamination

(for future version of document)

Pipe breaks

Backflow incident